# High-Precision Magnetic Angle Sensor IC

## 1. Features

- AEC-Q100 Grade0 qualified
- ISO26262 ASILB qualified
- High-precision non-contact absolute rotary angle position detection
- Simple magnetic circuit design
- Wide operating temperature range: -40 ℃ to 150 ℃
- Selectable output modes: analog output, PWM output, SAE J2716 compliant SENT output, 2-wire current-mode PSI5 (V2.3) output
- Programmable angle measurement range (up to 360°)
- Programmable linear transfer characteristics (custom 4-point / 8-point, or optional 16-segment / 32-segment equally spaced curves)
- 32-bit programmable user ID
- Differential Hall sensing for stray magnetic field immunity
- Rich on-chip diagnostic functions
- Overcurrent, overvoltage protection; undervoltage protection
- Package types: SOP8, TSSOP16, SIP3

## 2. Applications

- Non-contact absolute angle position detection
- Accelerator pedal position sensor
- Gear selector position detection
- Throttle Valve and EGR valve angular position detection
- Body height sensor
- Rotary switch angle position detection

## 3. Description

SC69411 is a differential Hall effect-based angular position sensor chip released by Semiment Technology. The chip integrates a fully differential Hall sensing matrix at the core, which detects corresponding sine and cosine position signals from an overhead S/N pole pair magnet. After being amplified by the preamplifier, the signals are sampled by the internal analog-to-digital conversion circuit. The chip's proprietary DSP circuit performs angle calculation, and finally outputs the absolute position information (0-360 degrees) of the magnet rotation through various interfaces.

SC69411 provides multiple output modes: analog output, PWM output, $I^2C$ output, digital SENT output, and 2-wire current-mode PSI5 output. The output curve can be programmed via custom 4-point / 8-point mapping or optional 16-/32-segment equally spaced linearization.

SC69411 is suitable for non-contact on-axis position detection, and is ideal for automotive applications such as accelerator pedals, electronic throttles, EGR valves, gear selectors, and body height systems.

The chip is available in three package types: SOP8, TSSOP16 and SIP3. The lead frame features matte tin plating, and the molding compound adopts halogen-free green material, fully complying with environmental protection and halogen-free requirements.

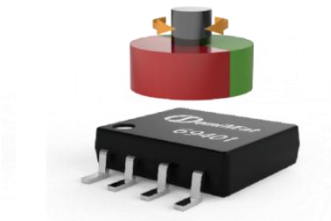


Fig.1 Product schematic diagram

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# 4. Terminal Configuration

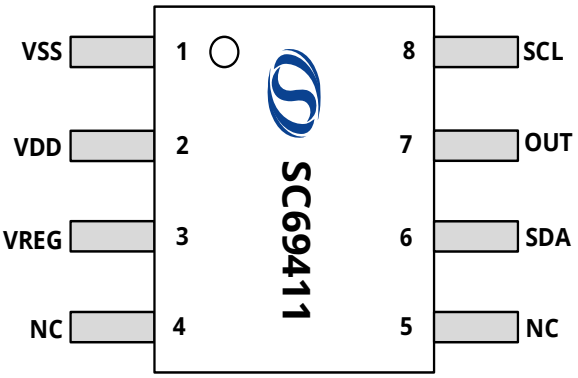



Fig.2 SOP8 Pin Description

| Terminal | | Type | Description |
|---|---|---|---|
| Name | Number | | |
| VSS | 1 | GND | Ground |
| VDD | 2 | Power | Power supply input |
| VREG | 3 | Power Output | Internal power supply |
| NC | 4 | - | Connect to GND |
| NC | 5 | - | Connect to GND |
| SDA | 6 | Test | Test pin, connect to GND |
| OUT | 7 | Output / Digital Input | Analog output; PWM; SENT |
| SCL | 8 | Test | Test pin, connect to GND |

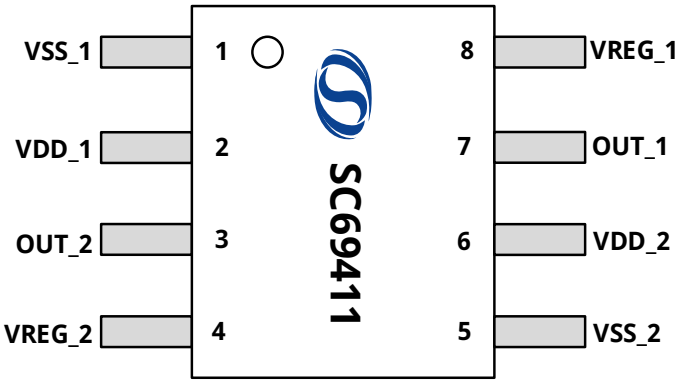



Fig.3 Dual-channel SOP8 Pin Description

| Terminal | | Type | Description |
|---|---|---|---|
| Name | Number | | |
| VSS_1 | 1 | GND | Channel 1 -- ground |
| VDD_1 | 2 | Power | Channel 1 -- power supply input |
| OUT_2 | 3 | Output / Digital Input | Channel 2 -- analog output; PWM; SENT |
| VREG_2 | 4 | Power Output | Channel 2 -- internal power supply |
| VSS_2 | 5 | GND | Channel 2 -- ground |
| VDD_2 | 6 | Power | Channel 2 -- power supply input |
| OUT_1 | 7 | Output / Digital Input | Channel 1 -- analog output; PWM; SENT |
| VREG_1 | 8 | Power Output | Channel 1 -- internal power supply |

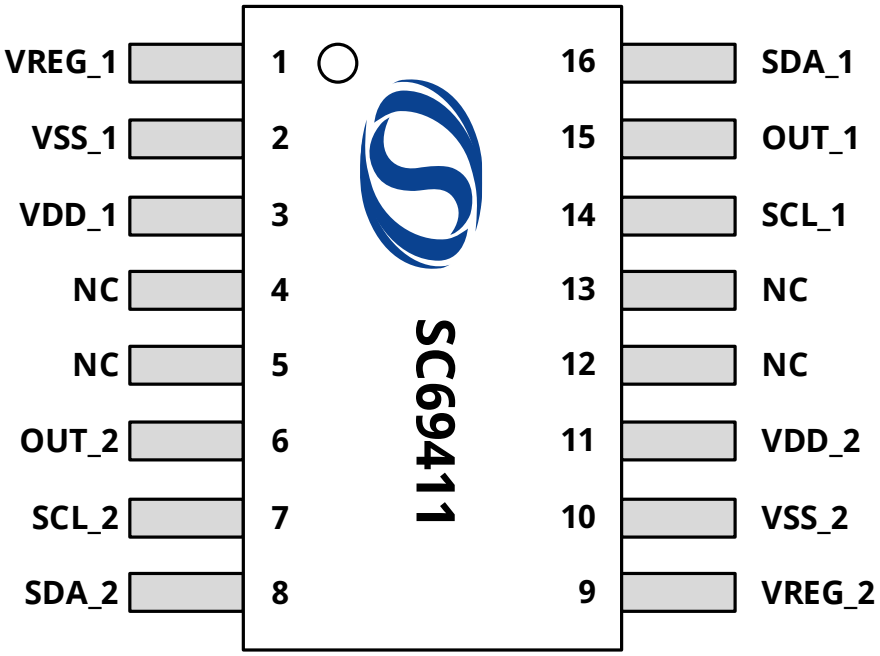



Fig.4 TSSOP16 Pin Description

| Terminal | | Type | Description |
|---|---|---|---|
| Name | Number | | |
| VREG_1 | 1 | Power Output | Channel 1 -- internal power supply |
| VSS_1 | 2 | GND | Channel 1 -- ground |
| VDD_1 | 3 | Power | Channel 1 -- power supply input |
| NC | 4 | - | Channel 1 -- connect to GND |
| NC | 5 | - | Channel 2 -- connect to GND |
| OUT_2 | 6 | Output / Digital Input | Channel 2 -- analog output; PWM; SENT |
| SCL_2 | 7 | Test | Channel 2 -- test pin, connect to GND |
| SDA_2 | 8 | Test | Channel 2 -- test pin, connect to GND |
| VREG_2 | 9 | Power Output | Channel 2 -- internal power supply |
| VSS_2 | 10 | GND | Channel 2 -- ground |
| VDD_2 | 11 | Power | Channel 2 -- power supply input |
| NC | 12 | - | Channel 2 -- connect to GND |
| NC | 13 | - | Channel 1 -- connect to GND |
| SCL_1 | 14 | Test | Channel 1 -- test pin, connect to GND |
| OUT_1 | 15 | Output / Digital Input | Channel 1 -- analog output; PWM; SENT |
| SDA_1 | 16 | Test | Channel 1 -- test pin, connect to GND |

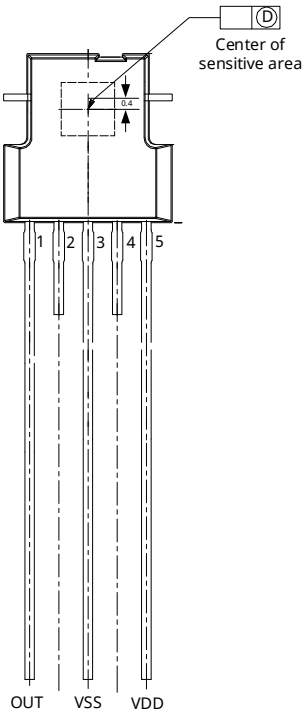


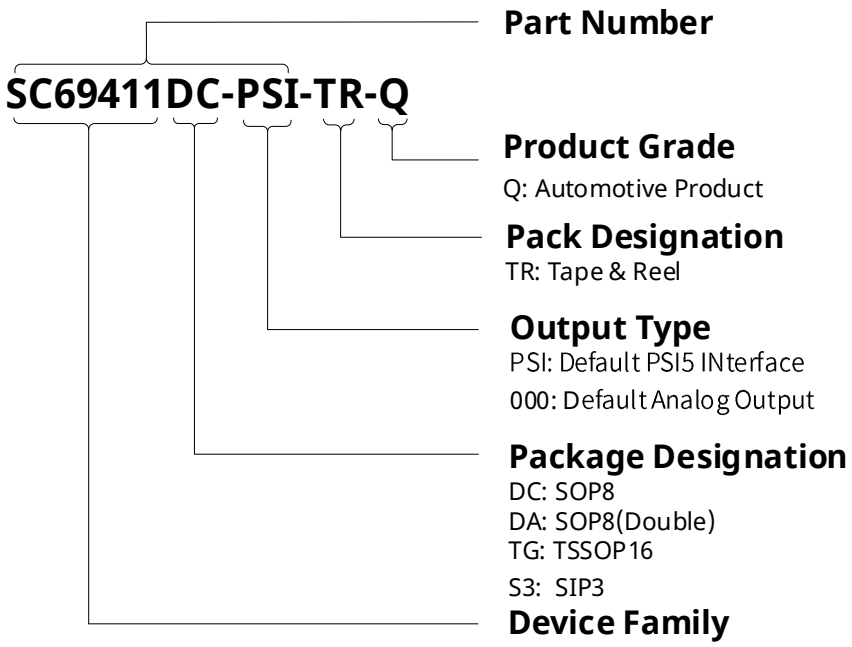

Fig.5 SIP3 Pin Description

| Terminal | | Type | Description |
|---|---|---|---|
| Name | Number | | |
| OUT | 1 | Output | Analog output; PWM; SENT |
| VSS | 3 | GND | Ground |
| VDD | 5 | Power Input | Power supply input / PSI5-OUT |

# 5. Ordering Information

| Ordering Information | Marking | Option | Grade | Ambient,$T_A$(℃) | Package | Packing | Quantity |
|---|---|---|---|---|---|---|---|
| SC69411DC-000-TR-Q | 69411 | Analog | Q | -40~150 | SOP8 | Tape&Reel | 4000pcs/reel |
| SC69411DC-PSI-TR-Q | 69411 | PSI | Q | -40~150 | SOP8 | Tape&Reel | 4000pcs/reel |
| SC69411DA-000-TR-Q | 69411 | Analog | Q | -40~150 | SOP8(D) | Tape&Reel | 4000pcs/reel |
| SC69411TG-000-TR-Q | 69411 | Analog | Q | -40~150 | TSSOP16 | Tape&Reel | 3000pcs/reel |
| SC69411S3-000-TR-Q | 69411 | Analog | Q | -40~150 | SIP3 | Tape&Reel | 2600pcs/reel |
| SC69411S3-PSI-TR-Q | 69411 | PSI | Q | -40~150 | SIP3 | Tape&Reel | 2600pcs/reel |

## Ordering Information Format



# 6. Absolute Maximum Ratings

Within the natural temperature range of operation (unless otherwise specified) (1)

| Symbol | Parameter | Test Conditions | Min. | Max. | Units |
|---|---|---|---|---|---|
| $V_{DD}$ | Supply Voltage | t<48h | -14 | 28 | V |
| | | t<60s | -18 | 37 | V |
| $V_{OUT}$ | Analog output voltage | t<48h | -10 | 24 | V |
| | | t<60s | -10 | 30 | V |
| $I_R$ | Reverse Output Voltage | - | - | 40 | mA |
| $T_A$ | Reverse Output Current | - | -40 | 150 | ℃ |
| $T_S$ | Operating Temperature | - | -65 | 165 | ℃ |
| H | Storage Temperature | - | -1 | 1 | T |

*Note:*

*(1) Stresses above those listed here may cause permanent damage to the device. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.*

# 7. ESD Protection

| Symbol | Parameter | Test Conditons | Min. | Max. | Units |
|---|---|---|---|---|---|
| $V_{ESD_HBM}$ | HBM | according to standard AEC-Q100-002 HBM, R=1.5kΩ, C=100pF | -6 | 6 | kV |
| $V_{ESD_CDM}$ | CDM | according to standard AEC-Q100-011C CDM | -750 | 750 | V |

# 8. Operating Characteristics

## Electrical Characteristics

| Symbol | Parameters | Test Conditions | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|---|
| $V_{DD}$ | Supply voltage | 5V mode | 4.5 | 5.0 | 5.5 | V |
| $V_{DD_3.3V}$ | | $V_{REG}$ and $V_{DD}$ are connected to 3.3V at the same time, $U_{VLO_3P3EN}$=1 | 3.15 | 3.3 | 3.6 | V |
| $V_{DD}$ | | PSI5 | 4.1 | - | 12 | V |
| $I_{DD}$ | Supply current | Single path SOP8 | - | 8 | 10 | mA |
| $I_{surge}$ | Isurge current | Single path SOP8 | - | - | 25 | mA |
| $I_{OCP}$ | Overcurrent alarm | Single path SOP8 | - | 25 | 35 | mA |
| $I_{ERRSTART}$ | Starting current error | PSI5 | -2 | - | 2 | mA |
| $\Delta_{IS}$ | Sink current | PSI5-Normal power mode | 22 | 26 | 30 | mA |
| | | PSI5-Low power mode | 11 | 13 | 15 | mA |
| $IS_{DRIFT}$ | Static current offset | PSI5 | -4 | - | 4 | mA |
| $V_{REG}$ | Regulated voltage | 5V mode | 3.13 | 3.3 | 3.47 | V |
| $V_{REGOVP}$ | Regulated voltage overdrive detection | 5V mode | 3.5 | 3.7 | 3.89 | V |
| $V_{REGUVL}$ | Regulated voltage low detection | 5V mode | 2.8 | 2.9 | 3.0 | V |
| $V_{DDstartH}$ | Starting voltage | 5V mode | 3.8 | - | - | V |
| $V_{DDstartHyst}$ | Starting voltage hysteresis | 5V mode | - | 100 | - | mV |
| $V_{UVLO}$ | Undervoltage detection voltage | $U_{VLO_3P3}$=0 | - | 3.9 | - | V |
| | | $U_{VLO_3P3}$=1 | - | 2.8 | - | V |
| $V_{UVLOHYS}$ | Undervoltage detection hysteresis | - | 50 | 100 | 200 | mV |
| $V_{OVP}$ | Overvoltage protection voltage | - | 6.0 | 6.2 | 6.4 | V |
| | | PSI5 | 22 | 24 | 26 | V |
| $V_{OVPHYST}$ | Overvoltage detection hysteresis | - | 50 | - | 200 | mV |
| | | PSI5 | 0.8 | 1.4 | 2.0 | V |
| $I_{short}$ | Output short-circuit current | Shorted to ground, analog output | - | - | 15 | mA |
| | | Shorted to ground, PWM, SENT push-pull outputs | - | - | 30 | mA |
| | | Shorted to power, analog output | - | - | 15 | mA |
| | | Short-Circuit to power, PWM, SENT push-pull outputs | - | - | 30 | mA |
| $R_L$ | Analog output load resistance | Pull-up resistor, connected to power supply | 4.7 | 10 | - | kΩ |
| | | Pull-down resistor, connected to ground | 4.7 | 10 | - | kΩ |
| $R_{L_PWM}$ | PWM output load resistance | Pull-up resistor, connected to power supply | 1 | - | - | kΩ |
| | | Pull-down resistor, connected to ground | 1 | - | - | kΩ |

## Electrical Characteristics (Continued)

| | | | | | | |
|---|---|---|---|---|---|---|
| $V_{sat_lo}$ | Analog output saturation level | Pull-up resistor R≥10k, connected to power supply | - | 0.5 | 2.0 | $\%V_{DD}$ |
| | | Pull-up resistor R≥4.7k, connected to power supply | - | 2.5 | 3.0 | $\%V_{DD}$ |
| $V_{sat_hi}$ | | Pull-down resistor R≥4.7k, connected to ground | 95 | 98 | - | $\%V_{DD}$ |
| | | Pull-down resistor R≥10k, connected to ground | 97 | 99 | - | $\%V_{DD}$ |
| $V_{satD_lo}$ | Digital output level | Pull-up resistor R≥10k, connected to the power supply | - | 0.5 | 1.0 | $\%V_{DD}$ |
| | | Pull-up resistor R≥1k, connected to the power supply | - | 2.5 | 4.0 | $\%V_{DD}$ |
| $V_{satD_hi}$ | | Pull-down resistor R≥1k, connected to ground | 85 | 90 | - | $\%V_{DD}$ |
| | | Pull-down resistor R≥10k, connected to ground | 97.5 | 98.0 | - | $\%V_{DD}$ |
| $D_{sat_lo}$ | Active diagnostic output level | Pull-up resistor R≥10k | - | 0.5 | 1.0 | $\%V_{DD}$ |
| | | Pull-up resistor R≥4.7k | - | 1 | 2 | $\%V_{DD}$ |
| $D_{sat_hi}$ | | Pull-down resistor R≥10k | 97.5 | 98.5 | - | $\%V_{DD}$ |
| | | Pull-down resistor R≥4.7k | 95 | 97 | - | $\%V_{DD}$ |
| $BV_{SS}PD$ | Passive diagnostic output level (Open drain) | $V_{SS}$ Open, pull-down resistor, R≤10k | - | 0.5 | 1.6 | $\%V_{DD}$ |
| $BV_{SS}PU$ | | $V_{SS}$ Open, pull-up resistor, R≥1k, pull-up to 5V | 99.5 | 100 | - | $\%V_{DD}$ |
| $BV_{DD}PD$ | | $V_{DD}$ Open, pull-down resistor, R≥1k | - | 0 | 0.5 | $\%V_{DD}$ |
| $BV_{DD}PU$ | | $V_{DD}$ Open, pull-up resistor, R≤10k, pull-up to 5V | 97.0 | 99.5 | - | $\%V_{DD}$ |
| Clamp_lo | Programmable clamp voltage | Programmable | 0 | - | 100 | $\%V_{DD}$ |
| Clamp_hi | | Programmable | 0 | - | 100 | $\%V_{DD}$ |

*Note:*

*(1)Based on the description in the table above, the SC69411 can meet the output range settings for the typical application of Fig.6*

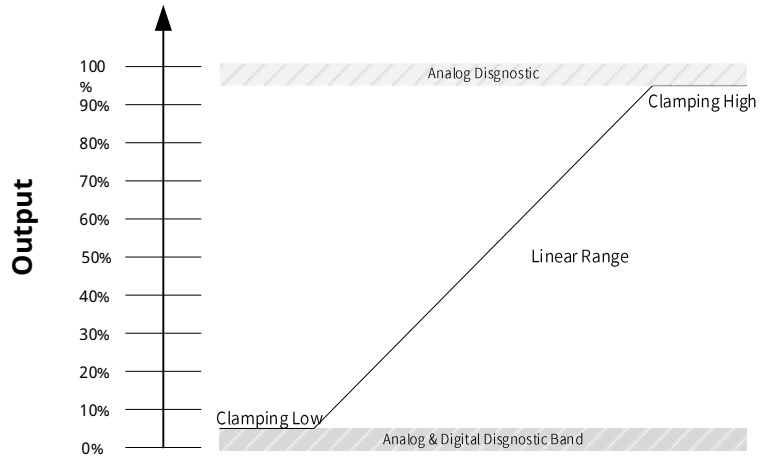



Fig.6 Example of Output Range for Typical Application

### Timing parameters - basic timing

| Symbol | Parameters | Test Conditions | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|---|
| $F_{CK}$ | Master Clock Frequency | Full Temperature Test | 7.8 | 8.2 | 8.5 | MHz |
| $\Delta F_{CK,T}$ | Main Clock Frequency Temperature Offset | - | -3 | - | 3 | $\%F_{CK}$ |
| Tper | Data Refresh Frequency | - | 121 | 128 | 134 | μs |
| Ts | Step Response Time | - | - | 128 | - | μs |
| $T_{POR}$ | Power-On Reset | - | - | 40 | - | μs |
| $T_{INIT}$ | Initialization Time | - | - | 3 | 5 | ms |
| SR | Simulated output conversion rate | $C_{OUT}$=10nF | - | 80 | - | V/ms |
| | | $C_{OUT}$=47nF | - | 85 | - | V/ms |
| | | $C_{OUT}$=100nF | - | 60 | - | V/ms |
| | | $C_{OUT}$=330nF | - | 20 | - | V/ms |

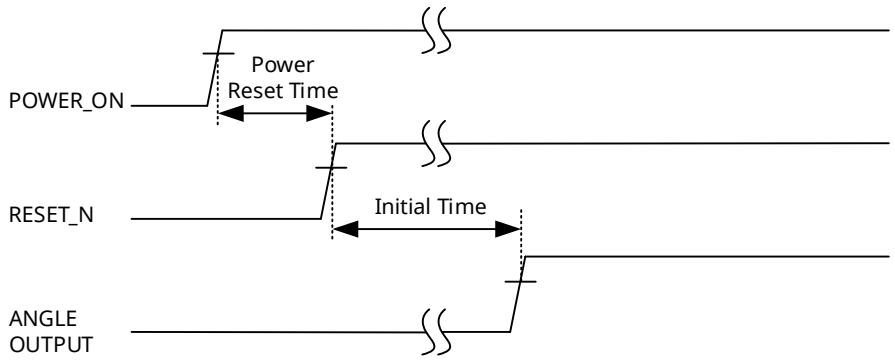



Fig.7 Power-on reset timing sequence

### Timing parameters - EEPROM Timing

| Symbol | Parameters | Test Conditions | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|---|
| $t_{PS}$ | Power-on Reset Time | | - | 100 | - | μs |
| $t_{PW}$ | | | - | 100 | - | μs |
| TIDLE | Standby Time | | - | 20 | - | ms |

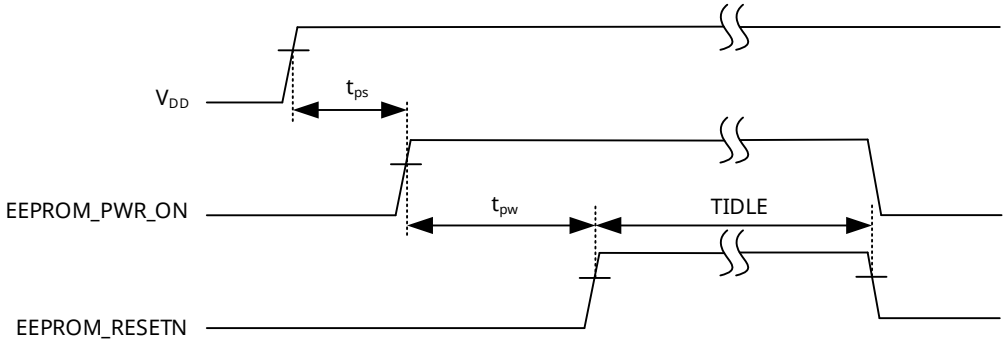



Fig.8 EEPROM Timing

## Timing parameters - PWM Output

| Symbol | Parameters | Test Conditions | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|---|
| $F_{PWM}$ | PWM Frequency | Frequency range | 100 | - | 2000 | Hz |
| $F_{PWM_Init}$ | PWM Frequency Initial Tolerances | 25℃ | - | - | ±2% | $F_{PWM}$ |
| | | 25℃ | - | - | ±1% | $F_{PWM}$ |
| $\Delta F_{PWM}$ | PWM Frequency Temperature Drift | PWM frequency temperature drift | - | - | ±3% | $F_{PWM}$ |
| $T_{rise_LSD}$ | PWM Output Rise Time (Open-Drain Output) | 4.7nF, $R_L$=1kΩ pull-up | - | 10 | - | μs |
| | | 10nF, $R_L$=1kΩ pull-up | - | 20 | - | μs |
| $T_{rise_PP}$ | PWM Output Rise Time (Push-Pull Output) | 4.7nF, $R_L$=1kΩ pull-up | - | 3 | - | μs |
| | | 10nF, $R_L$=1kΩ pull-up | - | 4 | - | μs |
| $T_{fall_LSD}$ | PWM Output Fall Time (Open-Drain Output) | 4.7nF, $R_L$=1kΩ pull-up | - | 2 | - | μs |
| | | 10nF, $R_L$=1kΩ pull-up | - | 4 | - | μs |
| $T_{fall_PP}$ | PWM Output Fall Time (Push-Pull Output) | 4.7nF, $R_L$=1kΩ pull-up | - | 2 | - | μs |
| | | 10nF, $R_L$=1kΩ pull-up | - | 4 | - | μs |

## Timing parameters - SENT Output

| Symbol | Parameters | Test Conditions | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|---|
| $TICK_{time}$ | Tick Time | - | 1.5 | - | 6 | μs |
| $N_{nibble}$ | Number of data Nibble | - | 3 | 6 | - | - |
| $T_{rise}$ | SENT edge rise time | Between 1.1V and 3.8V | - | 12.5 | 18 | μs |
| $T_{fall}$ | SENT edge fall time | Between 1.1V and 3.8V | - | 5.3 | 6.5 | μs |
| Npp | SENT frame cycle (no pause bit) | - | 154 | - | 270 | ticks |
| Ppc | SENT frame cycle (with pause bit) | - | 282 | - | 922 | ticks |
| A.1 | Sensor type | Dual Body Position Sensor | - | - | - | - |
| A.3 | | Single Reliable Sensor | - | - | - | - |
| $T_{FRAME}$ | SENT frame cycle (slow) | Standard data sequence (40 frames) | - | 691 | - | ms |
| | | Extended Data Sequence (24 frames) | - | 415 | - | ms |

### Timing parameters – PSI5 Output

| Symbol | Parameters | Test Conditions | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|---|
| $T_{cycle}$ | Communication cycle | - | - | 300 | - | μs |
| | | - | - | 500 | - | μs |
| | | - | - | 1000 | - | μs |
| $T_{bit}$ | bit time | 125 kbit/s | 7.6 | 8.0 | 8.4 | μs |
| | | 189 kbit/s | 5.0 | 5.3 | 6.0 | μs |
| $T_{SHOLD}$ | Synchronization pulse hold time | - | 9 | - | - | μs |
| $T_{RISE}$ | Current slope rise time | - | 0.33 | - | 1 | μs |
| TFALL | Current slope fall time | - | 0.33 | - | 1 | μs |
| MSR | Mark/Space Ratio | ($t_{fall}$, 80 - $t_{rise}$, 20) / $T_{Bit}$ , ($t_{fall}$, 20 - $t_{rise}$, 80) / $T_{Bit}$ | 47 | 50 | 53 | % |

### Accuracy Parameters - Analog Output

| Symbol | Parameters | Test Conditions | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|---|
| $R_{ADC}$ | ADC resolution | - | - | 15 | - | bits |
| $R_{DAC}$ | Analog output resolution | - | - | 12 | - | bits |
| INL | DAC integral nonlinearity error | - | - | 5 | - | LSB |
| DNL | DAC differential nonlinearity error | - | 0.05 | 1 | 3 | LSB |
| $\Delta E_{ang}$ | Angle offset error | - | -1 | - | 1 | Deg |
| $\Delta E_{L}$ | Non-linearity error | - | -1 | - | 1 | Deg |
| $\Delta E_{temp}$ | Angular temperature drift error | - | -0.5 | - | 0.5 | Deg |
| $\Delta E_{ratio}$ | Proportion output error | $4.5V \le V_{DD} \le 5.5V$ | -0.05 | 0 | 0.5 | $\%V_{DD}$ |
| $N_{pk-pk}$ | Output noise | - | - | 0.05 | 0.2 | Deg |

### Accuracy Parameters - PWM Output

| Symbol | Parameters | Test Conditions | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|---|
| RSP | PWM Output Resolution | - | - | 12 | - | bits |
| $J_{DC}$ | PWM % Duty Cycle Jitter (Open-Drain Output) | 125Hz,4.7nF, $R_L$=1kΩ, Resistor Pull-Up | - | ±0.003 | ±0.016 | %DC |
| | | 250Hz,4.7nF, $R_L$=1kΩ, Resistor Pull-Up | - | ±0.005 | ±0.02 | %DC |
| | | 500Hz,4.7nF, $R_L$=1kΩ,Resistor Pull-Up | - | ±0.009 | ±0.035 | %DC |
| | | 1000Hz,4.7nF, $R_L$=1kΩ,Resistor Pull-Up | - | ±0.003 | ±0.016 | %DC |
| | | 2000Hz,4.7nF, $R_L$=1kΩ,Resistor Pull-Up | - | ±0.005 | ±0.02 | %DC |
| $J_{DC}$ | PWM % Duty Cycle Jitter (Push-Pull Output) | 125Hz,4.7nF, $R_L$=1kΩ, Resistor Pull-Up | - | ±0.003 | ±0.016 | %DC |
| | | 250Hz,4.7nF, $R_L$=1kΩ, Resistor Pull-Up | - | ±0.005 | ±0.02 | %DC |
| | | 500Hz,4.7nF, $R_L$=1kΩ, Resistor Pull-Up | - | ±0.009 | ±0.035 | %DC |
| | | 1000Hz,4.7nF, $R_L$=1kΩ,Resistor Pull-Up | - | ±0.003 | ±0.016 | %DC |
| | | 2000Hz,4.7nF, $R_L$=1kΩ,Resistor Pull-Up | - | ±0.005 | ±0.02 | %DC |
| $J_{PWM}$ | PWM Frequency Jitter (Open-Drain Output) | 125Hz-2000Hz,4.7nF, $R_L$=1kΩ,Resistor Pull-Up | - | ±0.04 | ±0.15 | Hz |
| $J_{PWM}$ | PWM Frequency Jitter (Push-Pull Output) | 125Hz-2000Hz,4.7nF, $R_L$=1kΩ,Resistor Pull-Up | - | ±0.04 | ±0.15 | Hz |

### Magnetic parameter

| Symbol | Parameters | Test Conditions | Min. | Typ. | Max. | Units |
|---|---|---|---|---|---|---|
| $H_{EXT}$ | Magnetic field strength | - | 10 | - | 120 | mT |
| $D_{mag}$ | Magnet diameter | - | - | 6 | - | mm |
| $H_{mag}$ | Magnet thickness | - | - | 2.5 | - | mm |
| AG | Magnet To Chip Gap | - | 0.5 | - | 3 | mm |
| | Magnetic material | - | - | NdFeB 35 | - | - |

# 9. Block Diagram

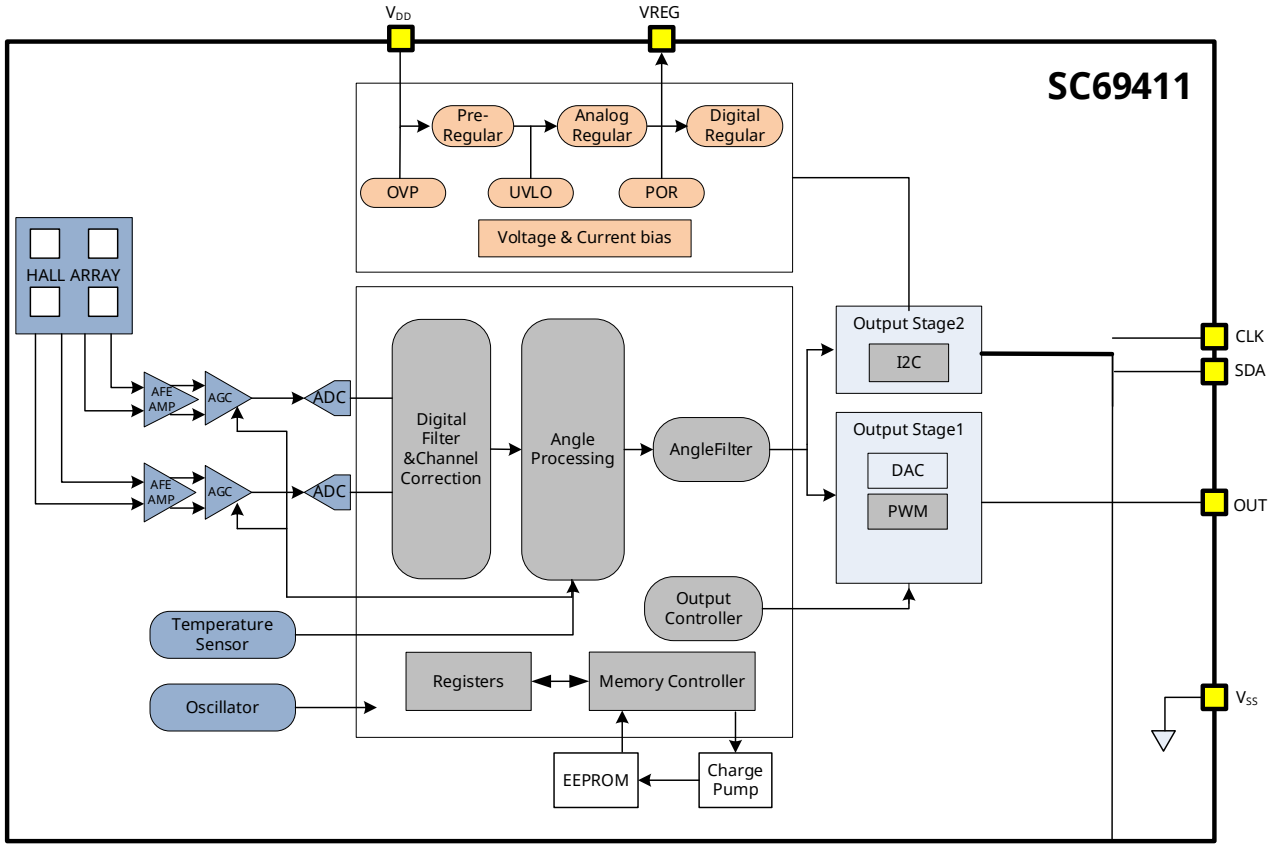



Fig.9 Functional block diagram

# 10. Function Description

### User programmable parameters

| Parameters | Description | Default | Bits |
|---|---|---|---|
| **General Parameter Settings** | | | |
| Filter | Filter selection | 2 | 2 |
| OUT_mode | Output mode | 0 | 3 |
| PWM_POL | PWM polarity | 0 | 1 |
| PWMT | PWM frequency | 0 | 16 |
| OUT_CONFIG | PWM，SENT output configuration | 0 | 2 |
| GAIN_G | Simulation of operational amplifier first-stage gain setting | 0 | 2 |
| AGC | Automatic Gain Control (AGC) for the 2nd stage of the analog op-amp | 1 | 1 |
| GAIN_F | Gain setting for the 2nd stage of the analog op-amp | 1 | 5 |
| USER_ID1 | User ID | 0 | 8 |
| USER_ID2 | User ID | 0 | 8 |
| USER_ID3 | User ID | 0 | 8 |
| USER_ID4 | User ID | 0 | 8 |
| EEPROM_LOCK_CODE | EEPROM lock code / EEPROM lock key | 0 | 8 |
| **SENT configuration parameters** | | | |
| SENT_FC_FORMAT | SENT frame data format selection | 1 | 1 |
| SENT_CH2_DATA | Channel 2 data configuration | 2 | 2 |
| SENT_INIT_DATA | SENT startup initialization data configuration | 0 | 1 |
| SENT_NIBBLE_NUMBER | Channel 2 data nibble count selection | 1 | 1 |
| SENT_TICK_TIME | SENT tick time configuration | 0 | 3 |
| SENT_LEGACY_CRC | CRC calculation options | 0 | 1 |
| SENT _SERIAL_CONFIG | Enhanced serial message selection | 1 | 1 |
| SENT_SLOW_EXTENDED | Serial message sequence selection | 0 | 1 |
| SENT_SLOW_BFIELD | Serial message extension sequence magnetic field configuration | 1 | 1 |
| SENT_PAUSE_OPTION | Selection of the measurement mode and the corresponding output protocol | 1 | 1 |
| SENT_REV | SENT version | 4 | 3 |
| SENT_MAN_CODE | factory code | 0 | 12 |
| SENT_FRAME_LENGTH | frame length | 297 | 10 |

## User programmable parameters(Continued)

| | | | |
|---|---|---|---|
| SENT_ SERIALERROR | Report diagnostics via fast channel | 0 | 3 |
| SENT_SENSOR_TYPE | Sensor type | 0x05 | 8 |
| SENT_CHANNEL_X1 | CHANNEL X1 | 0 | 12 |
| SENT_CHANNEL_X2 | CHANNEL X2 | 0 | 12 |
| SENT_CHANNEL_Y1 | CHANNEL Y1 | 0 | 12 |
| SENT_CHANNEL_Y2 | CHANNEL Y2 | 0 | 12 |
| SENT_SENSOR_ID1 | Sensor ID1 | 0 | 12 |
| SENT_SENSOR_ID2 | Sensor ID2 | 0 | 12 |
| SENT_SENSOR_ID3 | Sensor ID3 | 0 | 12 |
| SENT_SENSOR_ID4 | Sensor ID4 | 0 | 12 |
| SENT_OEM_CODE1 | OEM code 1 | 0 | 12 |
| SENT_OEM_CODE2 | OEM code 2 | 0 | 12 |
| SENT_OEM_CODE3 | OEM code 3 | 0 | 12 |
| SENT_OEM_CODE4 | OEM code 4 | 0 | 12 |
| SENT_OEM_CODE5 | OEM code 5 | 0 | 12 |
| SENT_OEM_CODE6 | OEM code 6 | 0 | 12 |
| SENT_OEM_CODE7 | OEM code 7 | 0 | 12 |
| SENT_OEM_CODE8 | OEM code 8 | 0 | 12 |
| **PSI5 configuration parameters** | | | |
| PSI5_OUT_MODE | PSI5 Output Mode | 3 | 2 |
| PSI5_PAYLOAD_SIZE | Effective data length | 20 | 5 |
| PSI5_ERROR_DETECTION | Error Detection Methods | 0 | 1 |
| PSI5_CYCLE_TIME | communications cycle | 0 | 2 |
| PSI5_TRANSMIT_SPEED | data transfer speed | 1 | 1 |
| PSI5_FORMAT_PRECEISION | Data frame format accuracy | 1 | 1 |
| PSI5_TRIGGER_LEVEL | Synchronized pulse trigger level setting | 1 | 1 |
| PSI5_TS[1..4]_ENABLE | Timeslot Enable | 0 | 1 |
| PSI5_TS[1..4]_SENSOR | Transmitted time slot data selection | 3 | 3 |
| PSI5_TS[1..4]_STARTTIME | time slot start time | 0 | 11 |
| PSI5_INIT_PHASES | Initialization Configuration | 0 | 2 |
| PSI5_INIT_I_DURATION | Initialization 1 phase cycle | 100 | 8 |

## User programmable parameters(Continued)

| | | | |
|---|---|---|---|
| PSI5_INIT_II_EXTRA_FIELDS | Initialization 2 Stage Extended Data Enable | 0 | 1 |
| PSI5_INIT_II_REPETITION | Initialization 2-stage repetition count | 1 | 2 |
| PSI5_INIT_METAINFO | PSI5 version | 6 | 4 |
| PSI5_INIT_INITLENGTH | Initialization 2-stage data length | 9 | 8 |
| PSI5_INIT_VENDORID | Supplier ID | 6 | 8 |
| PSI5_INIT_SENSORTYPE | Sensor type | 6 | 4 |
| PSI5_INIT_SENSORPARAMS | Sensor specified parameters | 0 | 8 |
| PSI5_INIT_SENSORCODE | Sensor Manufacturer Specified Parameters | 0 | 8 |
| PSI5_INIT_SENSORAPPCODE | Product version information | 0 | 12 |
| PSI5_INIT_PRODUCTIONDATE | date of manufacture | 0 | 16 |
| **Diagnostic Function Setting** | | | |
| DIAG_EN | Diagnostic Enable Bit | 1 | 1 |
| DIAG_MASK | Diagnostic Mask Register | 128 | 8 |
| DC_FAULT | Output duty cycle of PWM when supply voltage is too low | 0 | 8 |
| DC_FTL | Output duty cycle of PWM when magnetic field is too low | 0 | 8 |
| GAIN_THRESHOLD_LOW | Analog op amp second stage gain low thresholds | 0 | 5 |
| GAIN_THRESHOLD_HIGH | Analog op amp second stage gain high thresholds | 31 | 5 |
| FIELDTHOLD_LOW | low field strength threshold | 0 | 8 |
| FIELDTHOLD_HIGH | High field strength threshold | 255 | 8 |
| TEMPTHRESHOLD_LOW | low-temperature threshold | 0 | 7 |
| TEMPTHRESHOLD_HIGH | High-temperature threshold | 127 | 7 |
| DIAG_DEBOUNCE | Diagnostic de-jittering time | 0 | 3 |
| **Linear Transfer Curve Setting Parameters** | | | |
| OUTSLOPE_COLD | Low-temperature angle compensation coefficient | 0 | 8 |
| OUTSLOPE_HOT | High-temperature angle compensation coefficient | 0 | 8 |
| CLAMP_HIGH | Output high clamp | 65535 | 16 |
| CLAMP_LOW | Output low clamp | 0 | 16 |
| DP | Breakpoint / Zero point | 0 | 16 |
| CW | Rotation direction | 0 | 1 |
| WORK_RANGE_GAIN | 16-point / 32-point calibration working angle range (deg) | 360 | 16 |
| LNR_POINTS | Calibration point selection | 3 | 2 |

**User programmable parameters(Continued)**

| | | | |
|---|---|---|---|
| LNR_A_X | 4-point calibration , X-axis coordinates (angle) | 0 | 16 |
| LNR_B_X | | 0 | 16 |
| LNR_C_X | | 0 | 16 |
| LNR_D_X | | 0 | 16 |
| LNR_A_Y | 4-point calibration, Y-axis coordinates (%VDD ) | 0 | 16 |
| LNR_B_Y | | 0 | 16 |
| LNR_C_Y | | 0 | 16 |
| LNR_D_Y | | 0 | 16 |
| LNR_A_S | 4-point calibration, slope of each segment | 0 | 16 |
| LNR_B_S | | 0 | 16 |
| LNR_C_S | | 0 | 16 |
| LNR_D_S | | 0 | 16 |
| LNR4_S0 | 4-point calibration, initial slope | 0 | 16 |
| LNR4_Y5 | 4-point calibration, endpoint Y coordinate | 0 | 16 |
| LNR_Y0 | 4-point, 16-point/32-point calibration of initial point Y-coordinate | 0 | 16 |
| LNR9_Yn | 8-point calibration, Y-axis coordinates (n=0~8) | 0 | 9x16 |
| LNR9_Xn | 8-point calibration, X-axis coordinates (n=0~8) | 0 | 9x16 |
| LNR17_Yn | 16-point calibration, Y-axis coordinates (n=0~15) | 0 | 17x16 |
| LNR_DELTAYn | 32-point calibration, Y-axis coordinates (offset %) (n=0~31) | 0 | 32x8 |
| LNR_DELTA_Y_EXPAND | 32-point calibration, Y-axis coordinate deviation range setting | 3 | 2 |

## Output Modes

SC69411 supports five output modes: ratiometric analog output, PWM output, SENT output, I²C output and PSI5 output. PWM and SENT support both open-drain and push-pull output configurations.

| Parameters | Value | Description |
|---|---|---|
| OUT mode【2:0】+ | 0 | I²C output |
| | 1 | Analog output |
| | 2 | SENT |
| | 3 | PWM |
| | 4 | PSI5 |
| | 5 | I²C output |
| | 6 | I²C output |
| | 7 | I²C output |

## PWM Output Mode - Output Polarity Setting

| Parameters | Value | Description |
|---|---|---|
| PWM POL | 0 | Active low |
| | 1 | Active high |

## PWM Output Mode - Output frequency Setting

| Parameters | Value | Description |
|---|---|---|
| PWMT【15:0】 | 2000<br>40000 | 100-2000Hz frequency |

## PWM Output Mode – SENT Setting

| Parameters | Value | Description |
|---|---|---|
| OUT_CONFIG【1:0】 | 0 | Digital high-impedance output |
| | 1 | Digital output NMOS Open Drain |
| | 2 | Digital output PMOS Open Drain |
| | 3 | Digital push-pull output |

## PWM Output Mode – output waveform

PWM is set to PWM_POL=0, PWMT=0x0FA0 (1000Hz), and the PWM output waveform when the output duty cycle is 0.0244%.

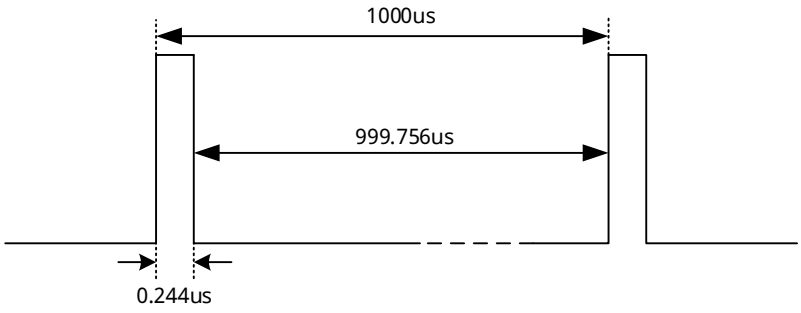



Fig.10 PWM output waveform when PWM_POL=0

PWM is set to PWM_POL=1, PWMT=0x0FA0 (1000Hz), and the PWM output waveform when the output duty cycle is 0.0244%.

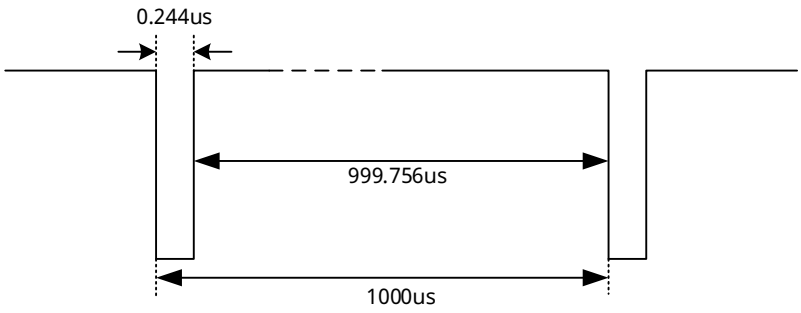



Fig.11 PWM output waveform when PWM_POL=1

## SENT protocol output

The digital SENT output of SC69411 is compatible with SAE J2716 APR2016.

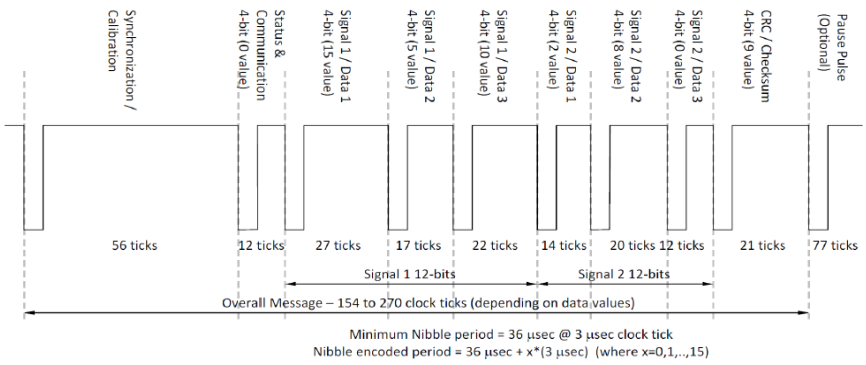



Fig.12 Example of 2-channel 12-bit data SENT information format

## SENT Protocol Output - Selection of SENT Fast Frame Data Format

| Parameters | Value | Description |
|---|---|---|
| SENT_FC_FORMAT【0】 | 0 | Dual Throttle Position Sensor |
| | 1 | Single safety position sensors |

## SENT Protocol Output - A.1 Dual Valve Body Position Sensor Data Frame

SC69411 sends a string of data bytes according to the dual valve body position sensor defined in Appendix A.1 of SAE J2716. CH1 continuously sends 12-bit sensor data, and the content of CH2 channel is set by SENT CH2 DATA.

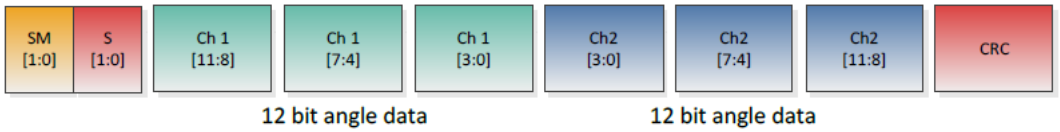



Fig.13 A.1 Dual Valve Body Position Sensor Data Frame

## SENT Protocol Output - Channel 2 Configuration for Number of Nibbles

| Parameters | Value | Description |
|---|---|---|
| SENT_NIBBLE_NUMBER【0】 | 0 | 3 nibbles |
| | 1 | 6 nibbles |

## SENT Protocol Output - Channel 2 Configuration Data

| Parameters | Value | Description |
|---|---|---|
| SENT CH2 DATA【1:0】 | 0 | 12-bit temperature sensor data |
| | 1 | 0xFF9(4089)-CH1 |
| | 2 | User-defined 12-bit data |
| | 3 | 0xFFF(4095)-CH1 |

## SENT Protocol Output - A.3 Single Reliable Position Sensor Data Frame

SC69411 transmits 12-bit sensor angle data of CH1 channel, roll counter, and the inverted data of the high 4 bits of angle data according to the single safety position sensor defined in Appendix A.3 of SAE J2716.




Fig.14 A.3 Single Reliable Position Sensor Data Frame

## SENT Protocol Output - Initialization Start Frame

| Parameters | Value | Description |
|---|---|---|
| SENT INIT DATA【1:0】 | 0 | OEM Requirements - 0xFFF |
| | 1 | SAE Compatible-0x000 |

## SENT Protocol Output - SENT tick Time Configuration

| Parameters | Value | Description |
|---|---|---|
| OUT mode【2:0】 | 0 | 3.0us Standard SENT |
| | 1 | Reserved |
| | 2 | Reserved |
| | 3 | 1.5us Fast SENT |
| | 4 | Reserved |
| | 5 | Reserved |
| | 6 | 6.0us Slow SENT |
| | 7 | Reserved |

## SENT Protocol Output - Frame Length Setting

| Parameters | Value | Description |
|---|---|---|
| SENT_FRAME_LENGTH【9:0】 | Default 297 ticks | Frame length setting |

## SENT Protocol Output - SENT Version Selection

| Parameters | Value | Description |
|---|---|---|
| SENT_REV【2:0】 | 0 | 没有指定版本 |
| | 1 | 2007 |
| | 2 | 2008 |
| | 3 | 2010 |
| | 4 | 2016 |

## SENT Protocol Output - Enhanced Serial Message

The enhanced serial message is a communication protocol that can transmit a large amount of data and information IDs. Serial data is transmitted through the 2nd and 3rd bits SM[1:0] of the status and communication Nibble. As shown in the figure, a serial information frame consists of 18 consecutive SENT data information, and all 18 frames must be successfully received (without errors such as calibration pulse changes, data byte CRC errors, etc.).

The enhanced serial message includes an enhanced serial information with 12 - bit data and 8 - bit information ID. SM[0] contains 6 - bit CRC, followed by 12 - bit data. The information content is defined by the 8 - bit information ID transmitted in the SM[1] channel. By default, a short sequence consisting of 24 data is transmitted. An extended sequence can be optionally used to transmit a cycle containing 40 data.

In addition, at the end of the sequence (short sequence or extended sequence), the standard value of the magnetic field detected by the sensor can be optionally returned.

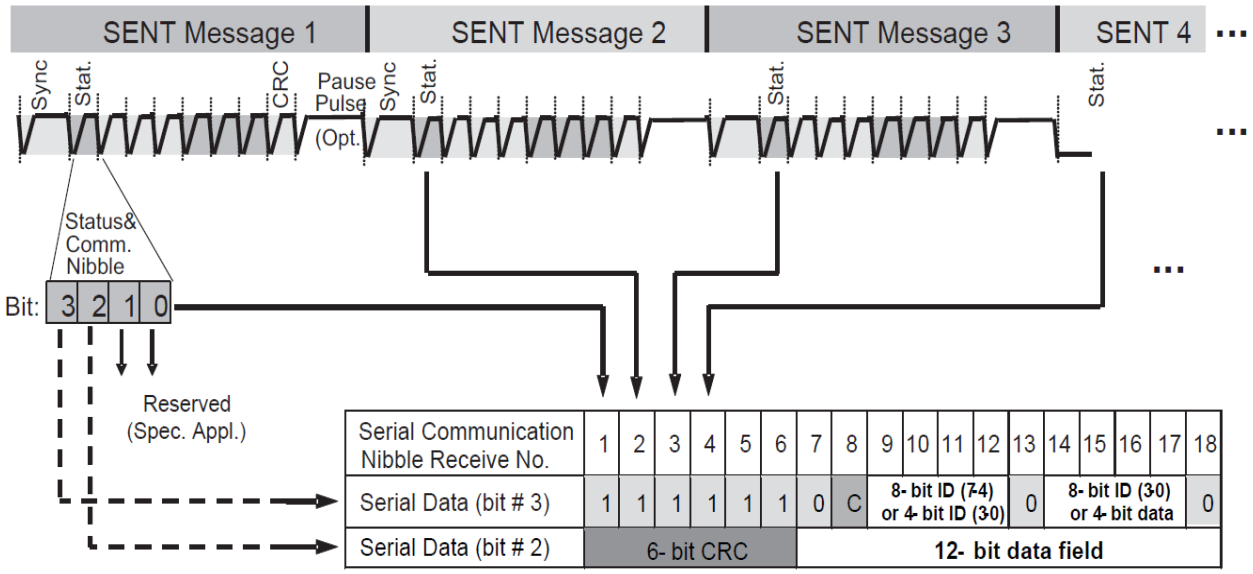



Fig.15 Enhanced Serial Message Data Format

| Num | 8-Bit Id | Element | Description |
|---|---|---|---|
| **Standard sequence** | | | |
| 1 | 0x01 | Diagnostic error code | Diagnostic register |
| 2 | 0x06 | SENT standard revision | SENT_REV |
| 3 | 0x01 | Diagnostic error code | Diagnostic register |
| 4 | 0x05 | Manufacturer code | SENT_MAN_CODE |
| 5 | 0x01 | Diagnostic error code | Diagnostic register |
| 6 | 0x03 | Channel 1 / 2 Sensor type | SENT_SENSOR_TYPE |
| 7 | 0x01 | Diagnostic error code | Diagnostic register |
| 8 | 0x07 | Fast channel 1: X1 | SENT_CHANNEL_X1 |
| 9 | 0x01 | Diagnostic error code | Diagnostic register |
| 10 | 0x08 | Fast channel 1: X2 | SENT_CHANNEL_X2 |
| 11 | 0x01 | Diagnostic error code | Diagnostic register |
| 12 | 0x09 | Fast channel 1: Y1 | SENT_CHANNEL_Y1 |
| 13 | 0x01 | Diagnostic error code | Diagnostic register |
| 14 | 0x0A | Fast channel 1: Y2 | SENT_CHANNEL_Y2 |
| 15 | 0x01 | Diagnostic error code | Diagnostic register |
| 16 | 0x23 | (Internal) temperature | Temperature register |
| 17 | 0x01 | Diagnostic error code | Diagnostic register |
| 18 | 0x29 | Sensor ID #1 | SENT_SENSOR_ID1 |
| 19 | 0x01 | Diagnostic error code | Diagnostic register |
| 20 | 0x2A | Sensor ID #2 | SENT_SENSOR_ID2 |
| 21 | 0x01 | Diagnostic error code | Diagnostic register |
| 22 | 0x2B | Sensor ID #3 | SENT_SENSOR_ID3 |
| 23 | 0x01 | Diagnostic error code | Diagnostic register |
| 24 | 0x2C | Sensor ID #4 | SENT_SENSOR_ID4 |
| **Extended sequence** | | | |
| 25 | 0x01 | Diagnostic error code | Diagnostic register |
| 26 | 0x90 | OEM Code #1 | SENT_OEM_CODE1 |
| 27 | 0x01 | Diagnostic error code | Diagnostic register |
| 28 | 0x91 | OEM Code #2 | SENT_OEM_CODE2 |
| 29 | 0x01 | Diagnostic error code | Diagnostic register |
| 30 | 0x92 | OEM Code #3 | SENT_OEM_CODE3 |
| 31 | 0x01 | Diagnostic error code | Diagnostic register |

| Num | 8-Bit Id | Element | Description |
|---|---|---|---|
| 32 | 0x93 | OEM Code #4 | SENT_OEM_CODE4 |
| 33 | 0x01 | Diagnostic error code | Diagnostic register |
| 34 | 0x94 | OEM Code #5 | SENT_OEM_CODE5 |
| 35 | 0x01 | Diagnostic error code | Diagnostic register |
| 36 | 0x95 | OEM Code #6 | SENT_OEM_CODE6 |
| 37 | 0x01 | Diagnostic error code | Diagnostic register |
| 38 | 0x96 | OEM Code #7 | SENT_OEM_CODE7 |
| 39 | 0x01 | Diagnostic error code | Diagnostic register |
| 40 | 0x97 | OEM Code #8 | SENT_OEM_CODE8 |
| **Magnetic field expansion** | | | |
| 41 | 0x80 | Field Strength | Magnetic Field Strength Register (Extended Sequence) |

*Note:*

*(1) SENT@ ID 23 = 8 * (T[C] – 35[C]) + 865 LSB12*

SENT Protocol Output - Enhanced Serial Message Slow Channel Selection

| Parameters | Value | Description |
|---|---|---|
| SERIAL_CONFIG【0】 | 0 | Disable Enhanced Serial Messaging Slow Channel |
| | 1 | Enable Enhanced Serial Message Slow Channel |

SENT Protocol Output - Enhanced Serial Message Sequence Selection

| Parameters | Value | Description |
|---|---|---|
| SENT_SLOW_EXTENDED【0】 | 0 | Short-sequence serial message selection |
| | 1 | Extended Sequence Serial Message Selection |

SENT Protocol Output - When the extended sequence is enabled, you can configure whether to use magnetic field extension

| Parameters | Value | Description |
|---|---|---|
| SENT_SLOW_BFIELD【0】 | 0 | No magnetic field expansion |
| | 1 | With magnetic field expansion |

SENT Protocol Output - PAUSE Configuration

| Parameters | Value | Description |
|---|---|---|
| SENT_PAUSE_OPTION【0】 | 0 | Continuous asynchronous angle acquisition, no pause bit in SENT |
| | 1 | Continuous synchronized angle acquisition, SENT with pause bit |

## PSI5 Protocol Output

According to the PSI5 protocol specification, information is transmitted by modulating the current of the power supply pin VDD. SC69411 complies with PSI5 protocol specification v2.3 and is backward compatible with v2.1 and v1.3.

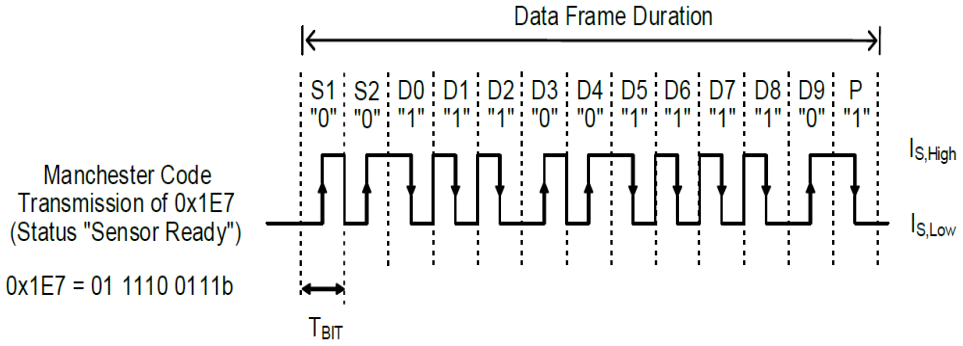



Fig.16 PSI5 Manchester Encoding Format

## PSI5 Protocol Output - Data Frame Format

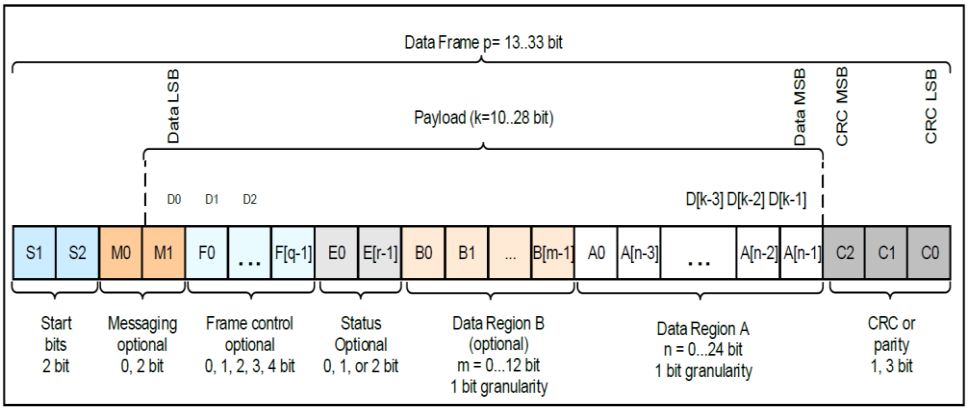



Fig.17 PSI5 Data Frame Format

## PSI5 Protocol Output - Error Detection Method

| Parameters | Value | Description |
|---|---|---|
| PSI5_ERROR_DETECTION【1:0】 | 0 | CRC mode |
| | 1 | Parity check |

## PSI5 Protocol Output - Data Length

| Parameters | Value | Description |
|---|---|---|
| PSI5_PAYLOAD_SIZE【4:0】 | 8-24 | Data length |
| PSI5_FORMAT_PRECEISION【0】 | 0 | Low precision |
| | 1 | Highly accurate |

| Parameters | Value | Description |
|---|---|---|
| PSI5_FORMAT_PRECEISION【0】 | 0 | Closing time slots |
| | 1 | Enable time slots |
| PSI5_TSx_SENSOR【2:0】 | 0 | Angular data |
| | 1 | - |
| | 2 | Temp |
| | 3 | Magnetic fields |
| | 4 | - |
| PSI5_TS[1..4]_STARTTIME 【10:0】 | 11 | Time slot start time |

## PSI5 Protocol Output - PSI5 Current Modulation Method

Data frames are transmitted through a current consumption modulator, and the encoding method is Manchester encoding. The low level (ISlow) represents the static current consumption of the sensor, and the high level (IShigh) is generated by the increase of the sensor's sink current (ISlow + ΔIShigh). The high level (IShigh) is generated by (ISlow + ΔIS). The sink current (ΔIS) and bit duration (TBIT) can be selected.

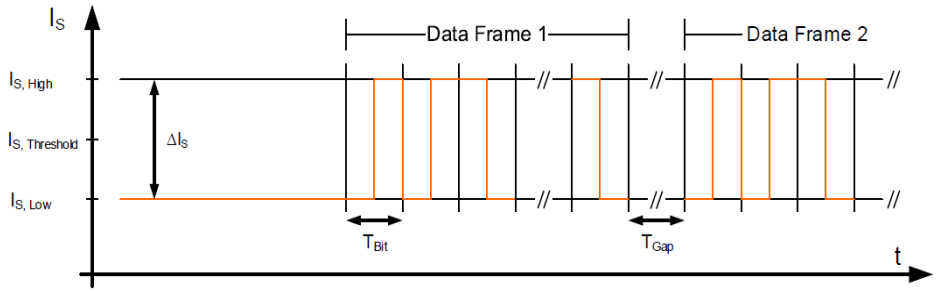



Fig.18 PSI5 data current modulation

| Parameters | Value | Description |
|---|---|---|
| PSI5_LOWCOMMON_MODE【1:0】 | 0 | Low current 13mA |
| | 1 | Normal current 26 mA |

## PSI5 Communication Mode

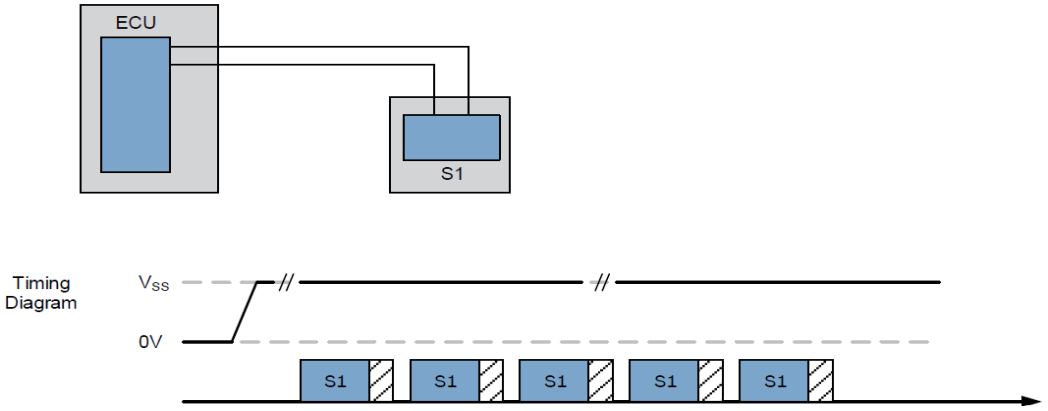



Fig.19 PSI5 Asynchronous mode

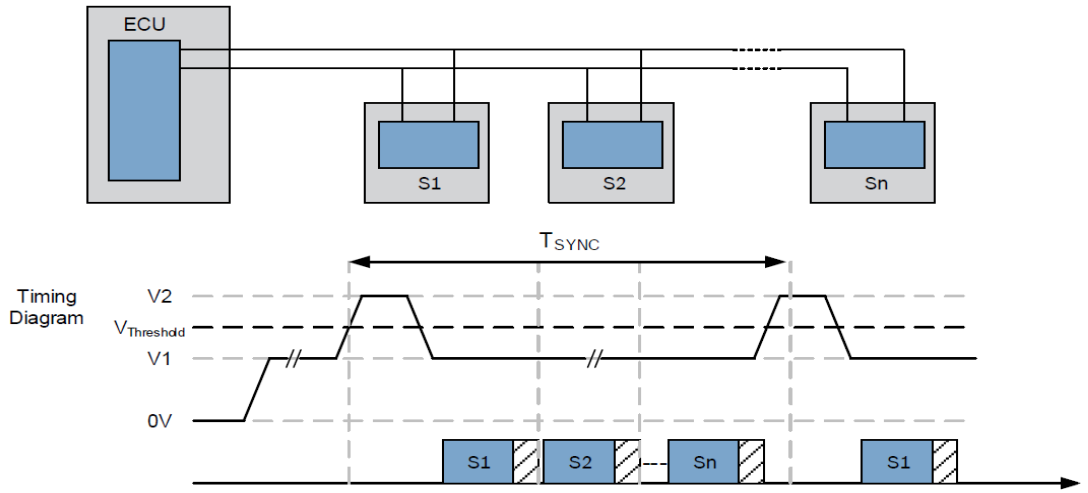



Fig.20 PSI5 Synchronous parallelism

## PSI5 Protocol Output - Output Mode

| Parameters | Value | Description |
|---|---|---|
| PSI5_OUT_MODE【1:0】 | 2 | Asynchronous mode |
| | 3 | Synchronous parallelism |

## PSI5 Protocol Output - Cycle Time

| Parameters | Value | Description |
|---|---|---|
| PSI5_OUT_MODE【1:0】 | 0 | 300μs |
| | 1 | 500μs |
| | 2 | 1000μs |

## PSI5 Protocol Output - Transmission Speed

| Parameters | Value | Description |
|---|---|---|
| PSI5_TRANSMIT_SPEED | 0 | Low speed 125 kbit/s |
| | 1 | High speed 189kbit/s |

## PSI5 Protocol Output - Setting of Synchronous Pulse Trigger Time

| Parameters | Value | Description |
|---|---|---|
| PSI5_TRIGGER_LEVEL【0】 | 0 | Reduced synchronization pulses |
| | 1 | Standard Synchronized Pulse |

## PSI5 Protocol Output - Sensor Initialization

| Initialization Phase 1 | Initialization Phase 2 | Initialization Phase 3 | Operating Mode |
|---|---|---|---|
| Activate (a plan) | Data content | State of affairs | Sensor or status data |

## PSI5 Protocol Output - Initialization Function Configuration

| Parameters | Value | Description |
|---|---|---|
| PSI5_INIT_PHASES【1:0】 | 0 | Normal initialization |
| | 1 | Close initialization phase 2 |
| | 2 | Close initialization phases 2 and 3 |
| | 3 | Reserved |

## PSI5 Protocol Output - Initialization Phase 1 Cycle

| Parameters | Value | Description |
|---|---|---|
| PSI5_INIT_I_DURATION【7:0】 | 100 | 50-200ms |

## PSI5 Protocol Output - Data content

| Parameters | Data field | Value | Description |
|---|---|---|---|
| Forced data | | | |
| 1 | F1(D1) | PSI5_INIT_METAINFO | PSI5 version |
| 2 | F2(D2，D3) | PSI5_INIT_INITLENGTH | Initialize data length |
| 3 | F3(D4，D5) | PSI5_INIT_VENDORID | Supplier ID |
| 4 | F4(D6，D7) | PSI5_INIT_SENSORTYPE | Sensor type |
| 5 | F5(D8，D9) | SI5_INIT_SENSORPARAMS | Sensor specified parameters |
| Extended data | | | |
| 6 | F6(D10，D11) | PSI5_INIT_SENSORCODE | Parameters specified by the sensor manufacturer |
| 7 | F7(D12-D14) | PSI5_INIT_SENSORAPPCODE | Product version information |
| 8 | F8(D15-D18) | PSI5_INIT_PRODUCTIONDATE | production date |
| 9 | F9(D19-D22) | SEMI_ID | SEMI_ID |

## PSI5 Protocol Output - Initialization Phase 3

| Parameters | Value | Description |
|---|---|---|
| “Sensor ready” | 0x1E7 | status data |
| “Sensor defect” | 0x1F4 | status data |

## Two-wire I2C protocol output (slave device)

### Sensor front-end setup

### First-level gain setting

| Parameters | Value | Description |
|---|---|---|
| GAIN_G【2:0】 | 0 | 2.5 |
| | 1 | 5 |
| | 2 | 10 |
| | 3 | 15 |

### Second-level gain setting

AGC enable is used to set the automatic gain control enable for the second-stage gain. When AGC is set to 1, the automatic gain control is enabled; when set to 0, the automatic gain control is disabled. If the AGC control bit is disabled, the second-stage gain is directly set through the register.

| Parameters | Value | Description |
|---|---|---|
| AGC | 0 | Digital high-impedance output |
| | 1 | Enable automatic gain control |
| GAIN_F【4:0】 | 0 | 1 |
| | 1 | 1.1 |
| | 2 | 1.21 |
| | ... | ... |
| | 29 | 15.86 |
| | 30 | 17.4 |
| | 31 | 20 |

### Traceable information

Each device includes a Saichuang factory ID and a user ID for traceability during factory shipment.

| Parameters | Value |
|---|---|
| USER_ID1【7:0】 | 0-255 |
| USER_ID2【7:0】 | 0-255 |
| USER_ID3【7:0】 | 0-255 |
| USER_ID4【7:0】 | 0-255 |

### EEPROM Write Protection

| Parameters | Value | Description |
|---|---|---|
| EEPROM_LOCK_ CODE【6:0】 | 0xAD | EEPROM read-only |
| | Other values | EEPROM can be read, written, and erased. |

## Diagnostic

| Diagnostic Enable Parameters | Value | Description |
|---|---|---|
| DIAG_EN | 0 | Close Diagnostics |
| | 1 | Enable Diagnostics |

### Diagnostic Mask Register

Corresponding to mask position 0, the fault will not trigger diagnosis; corresponding to mask position 1, the fault will trigger diagnosis.

| BIT7 | BIT6 | BIT5 | BIT4 | BIT3 | BIT2 | BIT1 | BIT0 |
|---|---|---|---|---|---|---|---|
| CRC Checksum Error | GAINF overflow | Current diagnosis | Digital Voltage Fault | CORDIC overflow | Fieldstreng thoverflow | ADC overflow | Temperature overflow |

### Analog Diagnostic Settings

| Parameters | Value | Description |
|---|---|---|
| DIAG_MO【1:0】 | 0 | Digital High Resistance State Output |
| | 1 | Digital Output NMOS OpenDrain |
| | 2 | Digital Output PMOS OpenDrain |
| | 3 | Digital push-pull output |

### PWM Diagnostic Settings

| Parameters | Value | Description |
|---|---|---|
| DC_FAULT【7:0】 | 0-255 | Output duty cycle of PWM when supply voltage is too low |
| DC_FTL【7:0】 | 0-255 | Output duty cycle of PWM when magnetic field is too low |

## Diagnostic Threshold

| 参数 | 数值 | 描述 |
|---|---|---|
| GAIN_THRESHOLD_LOW【4:0】 | 0-31 | Low threshold of gain for the second-stage operational amplifier |
| GAIN_THRESHOLD_HIGH【4:0】 | 0-31 | High threshold of gain for the second-stage operational amplifier |
| TEMPTHRESHOLD_LOW【6:0】 | 0-127 | low temperature threshold |
| TEMPTHRESHOLD_HIGH【6:0】 | 0-127 | high temperature threshold |
| FIELDTHOLD_LOW【7:0】 | 0-255 | Low threshold of field strength |
| FIELDTHOLD_HIGH【7:0】 | 0-255 | High threshold of field strength |

## Diagnostic debounce time setting

| Parameters | Value | STEP_UP TIME(ms) | STEP_DOWN TIME(ms) |
|---|---|---|---|
| DIAG_DEBOUNCE【2:0】 | 0 | 20 | 20 |
| | 1 | 20 | 30 |
| | 2 | 20 | 40 |
| | 3 | 40 | 40 |
| | 4 | 60 | 80 |
| | 5 | 80 | 100 |
| | 6 | 100 | 120 |
| | 7 | 120 | 140 |

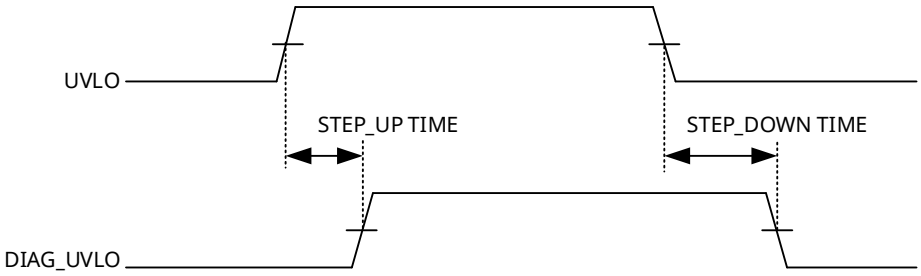



Fig.21 Diagnostic de-jittering timings

## Output parameter settings

### Angle output temperature compensation

The two parameters, OUTSLOPE_HOT and OUTSLOPE_COLD, are used to compensate for temperature-related offsets. This offset is applied to the angle before the clamping function. As shown in the following figure, the offset is calculated through the internal linearized temperature of the device.

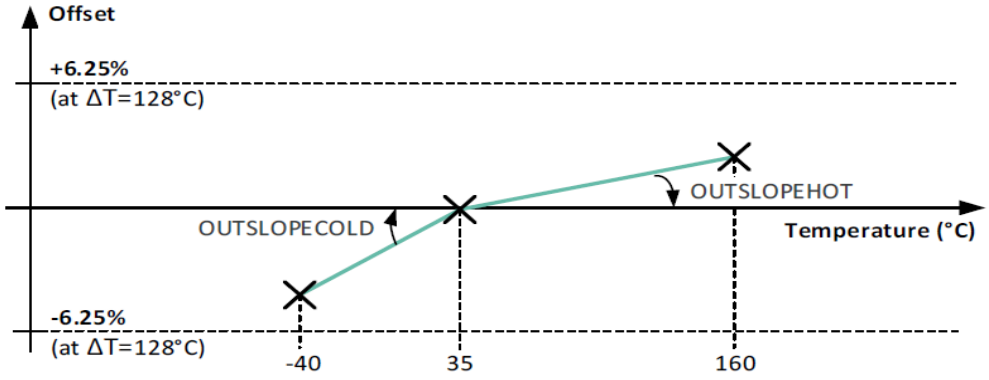



Fig.22 Temperature deviation map

The span of this offset is ±6.25% of the total output, and the linearized temperature depends on whether the temperature is below or above 35°C.

If the internal temperature of the chip is higher than 35℃： $Angle_{Compensated}$ = Angle − $\Delta T$ *OUTSLOPE_HOT /64

If the internal temperature of the chip is lower than 35℃： $Angle_{Compensated}$ = Angle − $\Delta T$ *OUTSLOPE_COLD /64

## Output Clamp Setting

The output clamp setting is used to limit the output voltage range.CLAMP_LOW sets the output voltage minimum and CLAMP_HIGH sets the output voltage maximum. Both parameters work for 4-point, 8-point, 16-point, and 32-point correction modes.

| Parameters | Value | Description |
|---|---|---|
| CLAMP_LOW | 0-100% | Low clamping |
| CLAMP_HIGH | 0-100% | High clamping |

### Breakpoint/Zero-DP

The breakpoint and zero point of SC69411 are the same point, which can be programmed at any point on the circumference. All angles are based on the breakpoint or zero point. DP is the transition point between 0 degrees and 360 degrees. For applications with a travel range less than 360 degrees, DP should not be set at the same position as the starting point of the working travel, but must be set outside the working travel.

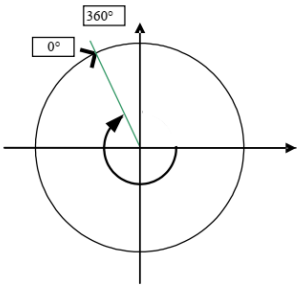



Fig.23 Schematic diagram of DP points

| Parameters | Value | Description |
|---|---|---|
| DP | 0-360 | Breakpoint/Zero (degrees) |

## Direction of rotation

The CW parameter defines the direction of rotation of the magnet.

Counterclockwise is defined as rotation in 1-4-5-8 pin order (SOP8 package) or 1-8-9-16 pin order (eTSSOP16L package); clockwise is defined as the opposite direction, rotation in 8-5-4-1 pin order (SOP8 package) or 16-9-8-1 pin order (TSSOP16L package).

| Parameters | Value | Description |
|---|---|---|
| CW | 0 | Counterclockwise |
| | 1 | Clockwise |

## 4-Point Calibration Mode

The SC69411 allows the user to divide the output curve through 4 points into up to 5 segments using the 4-point calibration mode, allowing the number of calibration points to be reduced to 2 or 3. The Y coordinate (-50%~100%) and X coordinate (0°~360°) of the 4 calibration points, and the slope of the 5 segments (LNR_S0,LNR_S1,LNR_S2,LNR_S3,LNR_S4) are fully set by the user. . To calculate the slope, two endpoints of the curve, 0 degree start and 360 degree end, are needed to calculate LNR_S0 and LNR_D_S.

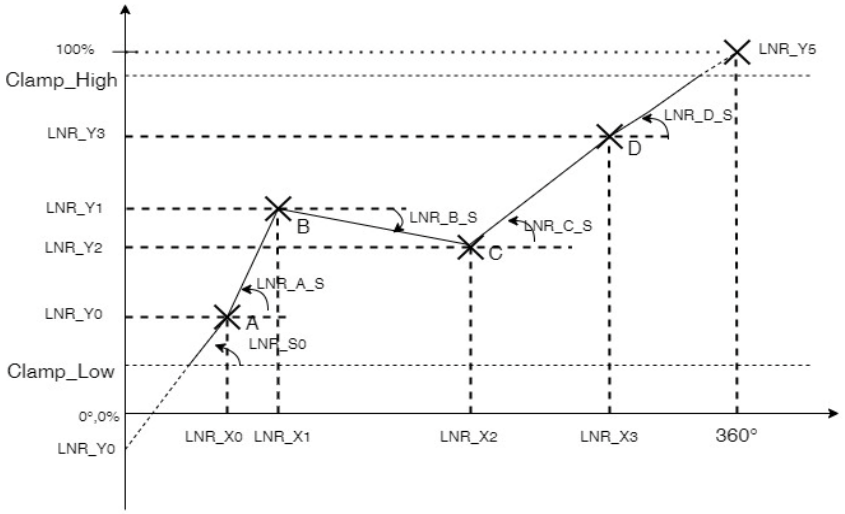



Fig.24 4-point calibration plot

## 8-Point Arbitrary Point Calibration Mode

The SC69411 allows the user to program the output curve desired by the user by customizing the X-coordinate (0°~ 360° ) and Y-coordinate (0%~100%) of any 8 calibration points. However, the slope cannot be set and can only be calculated from two neighboring points. A default fixed calibration point [0°, 0%] is also required as a starting point.

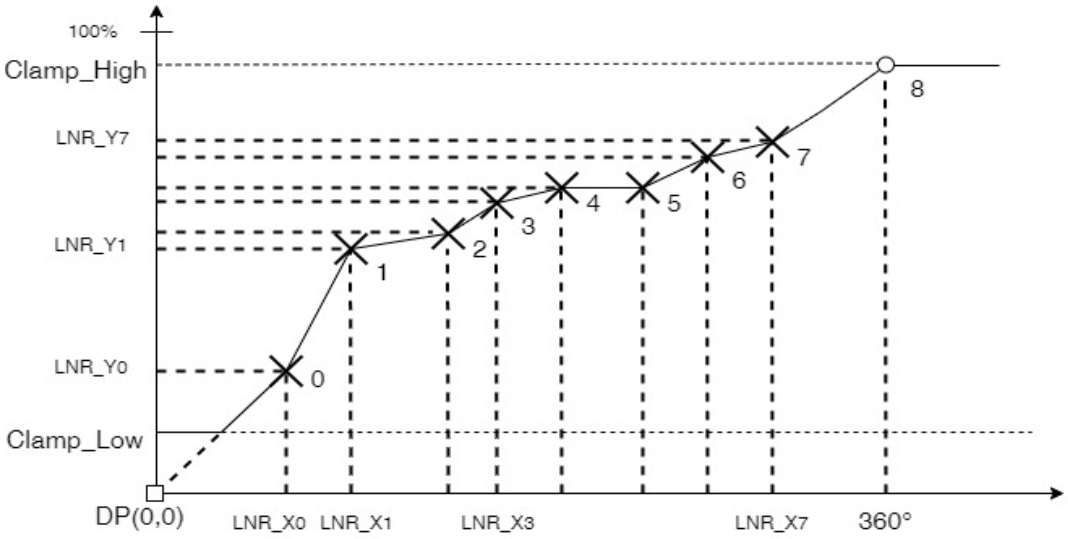



Fig.25 8-point calibration plot

## 16-Point Calibration Mode

The 16-point calibration mode allows the user to set only the Y-axis value of the coordinate point. x-axis coordinates are defined by the W value, which is divided into 16 segments within the WORK_RANGE range. y-point coordinates are allowed to range from -50% to +150% of the clamped voltage, which allows the clamped voltage to be in-between a certain segment (as shown in the following figure). But the output is still clamped voltage.

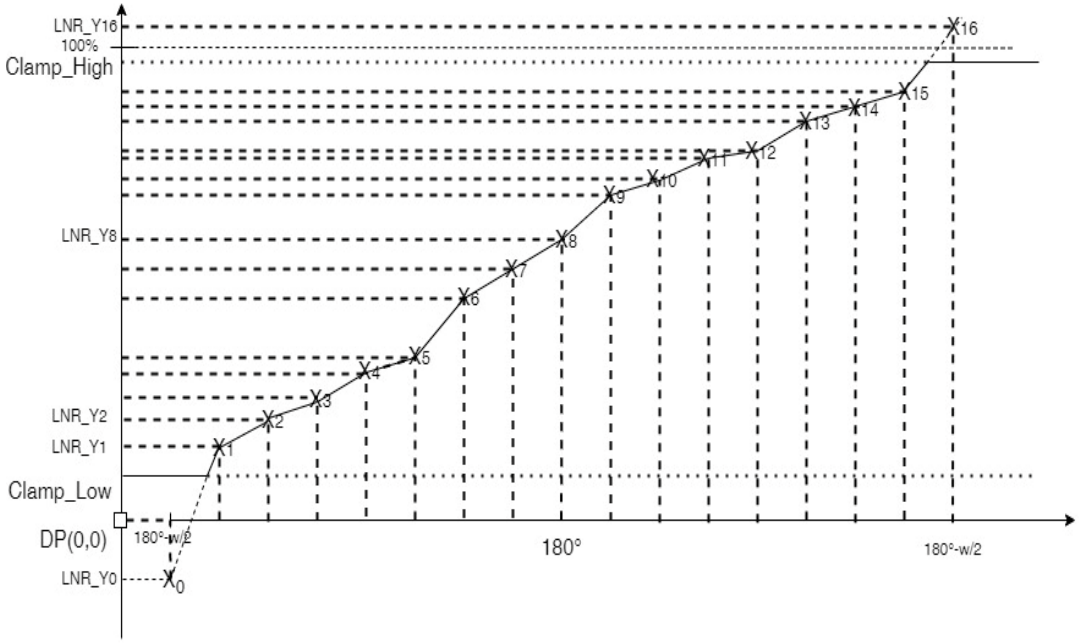



Fig.26 16-point calibration plot

## 32-point calibration mode

The work range W is defined by Workrange and is divided into 32 segments centered at 180°. The Y-axis coordinates consist of only 8 bits of data, so they are not absolute values but incremental coordinates. The two endpoints are (180°-w/2, 0%) and (180°+w/2,100%) to define an ideal curve, and ΔY is the fine-tuned value of Y corresponding to the X coordinate of the horizontal axis.

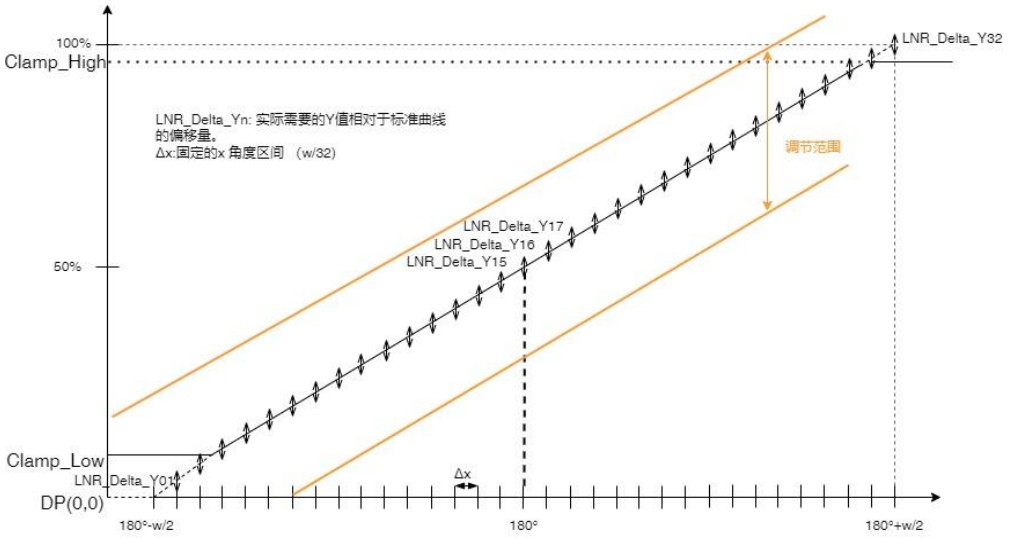



Fig.27 32-point calibration plot

32-point calibration, Y-axis coordinate deviation range setting

| Parameters | Value | Description |
|---|---|---|
| LNR_DELTA_Y_EXPAND【1:0】 | 0 | Tolerance range ±3.125% |
| | 1 | Tolerance range ±6.25% |
| | 2 | Tolerance range ±12.5% |
| | 3 | Tolerance range ±25% |

Angle Range Selection

$$w = \frac{WORK_RANGE_GAIN \times 360°}{0XFFFF}$$

Angular range $\theta_{min} = \frac{360°-w}{2}$ $\theta_{max} = \frac{360°+w}{2}$

θmin indicates the angle at 0% output, and θmax indicates the angle at 100% output. In order to correct the output properly, the angle range needs to be set to an integer multiple greater than 16.

Example of angular range setting:

| $WORK_RANGE_GAIN$ | $w$ (°) | $\theta_{min}$ (°) | $\theta_{max}$(°) | △X，16pts(°) | △X，32pts(°) |
|---|---|---|---|---|---|
| 0x1000 | 22.50034 | 168.7498 | 191.2502 | 1.406271 | 0.703136 |
| 0x1100 | 23.90661 | 168.0467 | 191.9533 | 1.494163 | 0.747082 |
| 0x1200 | 25.31289 | 167.3436 | 192.6564 | 1.582055 | 0.791028 |
| 0x1300 | 26.71916 | 166.6404 | 193.3596 | 1.669947 | 0.834974 |
| 0x2000 | 45.00069 | 157.4997 | 202.5003 | 2.812543 | 1.406271 |
| 0x2100 | 46.40696 | 156.7965 | 203.2035 | 2.900435 | 1.450217 |
| 0x2200 | 47.81323 | 156.0934 | 203.9066 | 2.988327 | 1.494163 |
| 0x2300 | 49.2195 | 155.3902 | 204.6098 | 3.076219 | 1.538109 |
| 0x3000 | 67.50103 | 146.2495 | 213.7505 | 4.218814 | 2.109407 |
| 0x3100 | 68.9073 | 145.5463 | 214.4537 | 4.306706 | 2.153353 |
| 0x3200 | 70.31357 | 144.8432 | 215.1568 | 4.394598 | 2.197299 |
| 0x3300 | 71.71984 | 144.1401 | 215.8599 | 4.48249 | 2.241245 |
| 0x4000 | 90.00137 | 134.9993 | 225.0007 | 5.625086 | 2.812543 |
| 0x4100 | 91.40764 | 134.2962 | 225.7038 | 5.712978 | 2.856489 |
| 0x4200 | 92.81392 | 133.593 | 226.407 | 5.80087 | 2.900435 |
| 0x4300 | 94.22019 | 132.8899 | 227.1101 | 5.888762 | 2.944381 |
| 0xFA00 | 351.5679 | 4.216068 | 355.7839 | 21.97299 | 10.9865 |
| 0xFB00 | 352.9741 | 3.512932 | 356.4871 | 22.06088 | 11.03044 |
| 0xFC00 | 354.3804 | 2.809796 | 357.1902 | 22.14878 | 11.07439 |
| 0xFD00 | 355.7867 | 2.106661 | 357.8933 | 22.23667 | 11.11833 |
| 0xFE00 | 357.193 | 1.403525 | 358.5965 | 22.32456 | 11.16228 |
| 0xFF00 | 358.5992 | 0.700389 | 359.2996 | 22.41245 | 11.20623 |
| 0xFFFF | 360 | 0 | 360 | 22.5 | 11.25 |

# 11. Typical Application Circuit

## Analog/PWM/SENT SOP8 Package Application Circuits

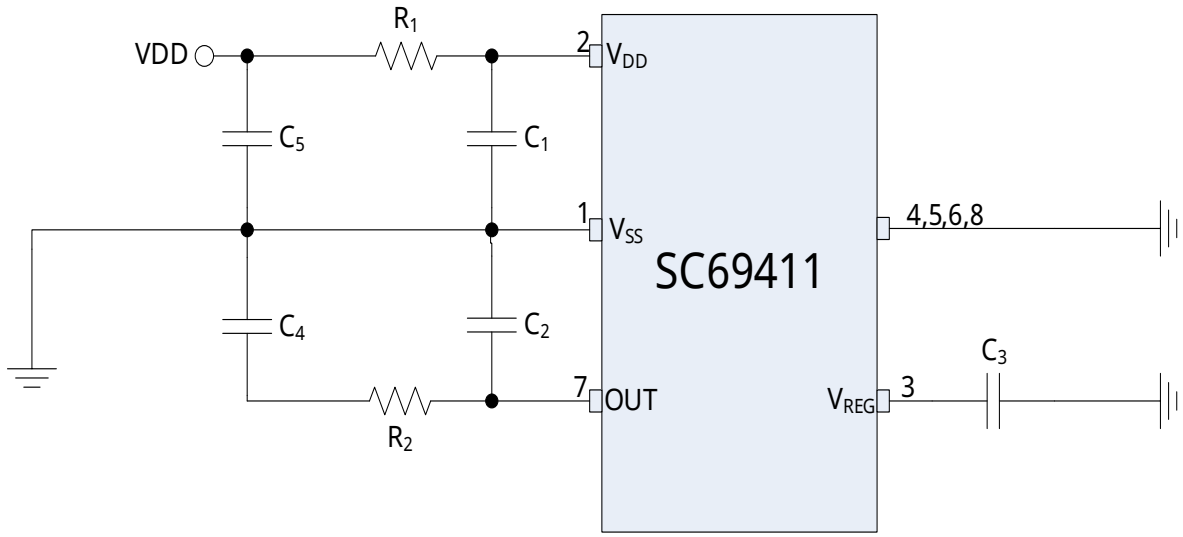



Fig.28 Analog/PWM/SENT SOP8 Reference Circuit Diagram

## Analog Output Reference

| Component | Min. | Typ. | Max. | Description |
|---|---|---|---|---|
| R1 | - | 0Ω | 10Ω | Reducing the impact of EMC increases the output error |
| R2 | - | 0Ω | 51Ω | Reduce the impact of EMC |
| C1 | 100nF | 100nF | - | Placement near pins |
| C2 | 47nF | 100nF | - | Placement near pins |
| C3 | 47nF | 100nF | 220nF | Placement near pins |
| C4 | - | 1nF | 10nF | Reduce the impact of EMC and place it close to the connector end |
| C5 | - | 1nF | 10nF | Reduce the impact of EMC and place it close to the connector end |

## Digital Output (PWM) Reference

| Component | Min. | Typ. | Max. | Description |
|---|---|---|---|---|
| R1 | - | 0Ω | 33Ω | Reducing the impact of EMC affects the output of high level |
| R2 | - | 0Ω | 51Ω | Reducing the impact of EMC affects the output of high and low levels |
| C1 | 100nF | 100nF | - | Placement near pins |
| C2 | 2.2nF | 4.7nF | 22nF | Placement near pins |
| C3 | 47nF | 100nF | 220nF | Placement near pins |
| C4 | - | 1nF | 10nF | Reduce the impact of EMC and place it close to the connector end |
| C5 | - | 1nF | 2.2nF | Reduce the impact of EMC and place it close to the connector end |

## Digital Output (SENT) Reference Value

| Component | Min. | Typ. | Max. | Description |
|---|---|---|---|---|
| R1 | - | 0Ω | 10Ω | Reducing the impact of EMC affects the output of high level |
| R2 | - | - | - | |
| C1 | 100nF | 100nF | - | Placement near pins |
| C2 | 1nF | 4.7nF | 22nF | Placement near pins |
| C3 | 47nF | 100nF | 220nF | Placement near pins |
| C4 | - | 1nF | 10nF | Reduce the impact of EMC and place it close to the connector end |
| C5 | - | - | 1nF | Reduce the impact of EMC and place it close to the connector end |

# Analog/PWM/SENT TSSOP16 Package Application Circuit

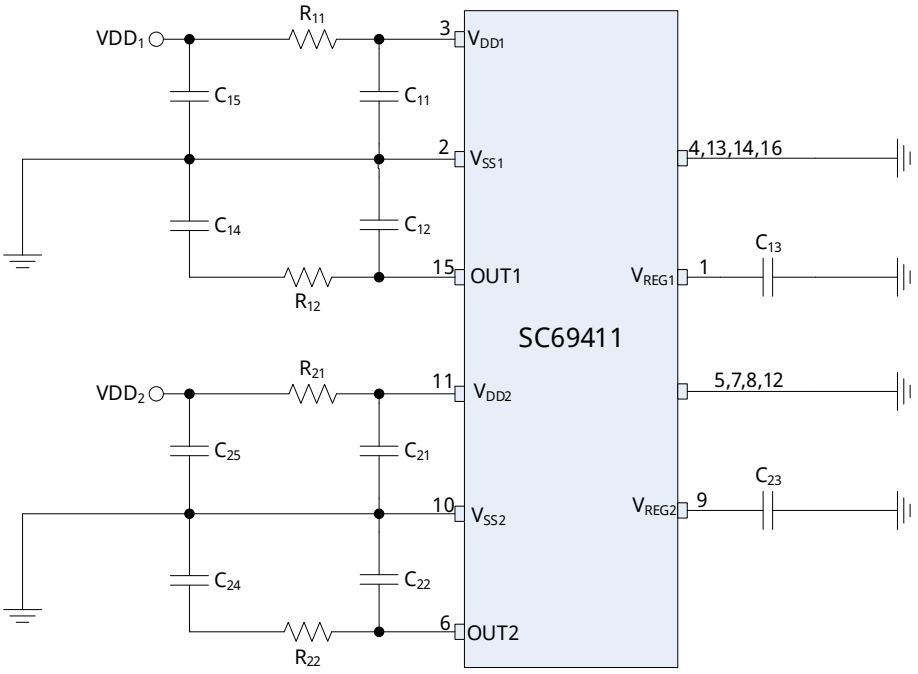



Fig.29 Analog/PWM/SENT TSSOP16 Reference Circuit Diagram

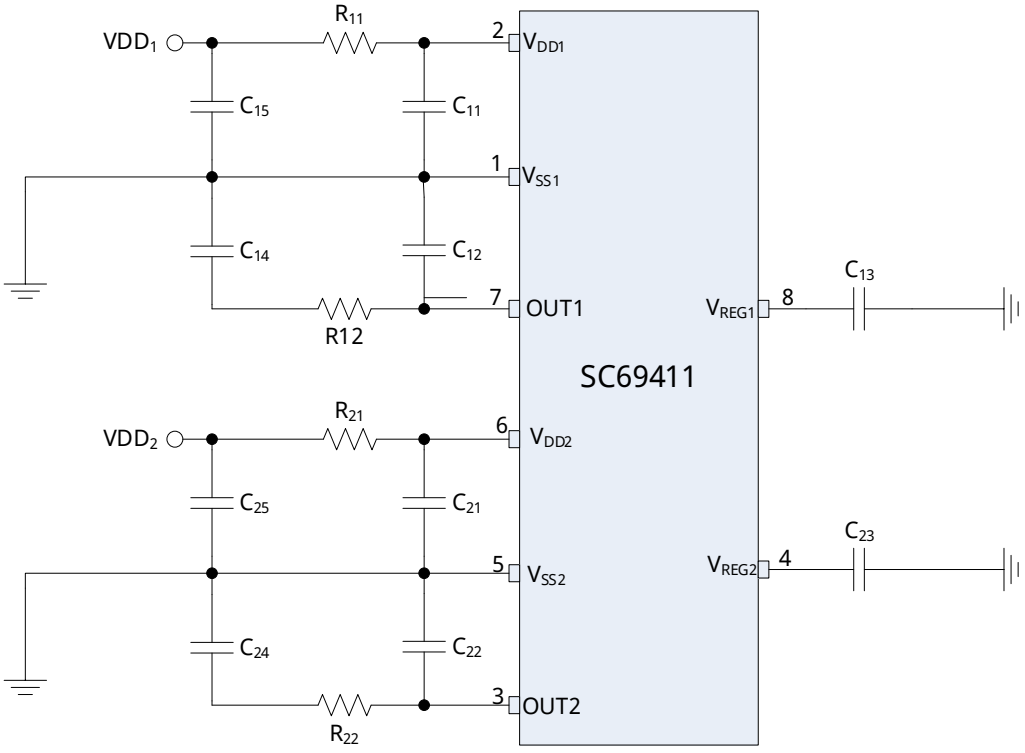



Fig.30 Analog/PWM/SENT SOP8 Reference Circuit Diagram

## Analog Output Reference Value

| Component | Min. | Typ. | Max. | Description |
|---|---|---|---|---|
| R11, R21 | - | 0Ω | 10Ω | Reduces EMC effects and increases measurement errors |
| R12, R22 | - | 0Ω | 51Ω | Reduces EMC effects and increases measurement errors |
| C11, C21 | 100nF | 100nF | - | Placement near pins |
| C12, C22 | 47nF | 100nF | - | Placement near pins |
| C13, C23 | 47nF | 100nF | 220nF | Placement near pins |
| C14, C24 | - | 1nF | 10nF | Reduced EMC impact, placed close to the connector end |
| C15, C25 | - | 1nF | 10nF | Reduced EMC impact, placed close to the connector end |

## Digital Output (PWM) Referencev Value

| Component | Min. | Typ. | Max. | Description |
|---|---|---|---|---|
| $R_{11}$, $R_{21}$ | - | 0Ω | 33Ω | Reduces EMC effects affecting the output high level |
| $R_{12}$, $R_{22}$ | - | 0Ω | 51Ω | Reduces EMC effects, affecting output high and low levels |
| $C_{11}$, $C_{21}$ | 100nF | 100nF | - | Placement near pins |
| $C_{12}$, $C_{22}$ | 2.2nF | 4.7nF | 22nF | Placement near pins |
| $C_{13}$, $C_{23}$ | 47nF | 100nF | 220nF | Placement near pins |
| $C_{14}$, $C_{24}$ | - | 1nF | 10nF | Reduced EMC impact, placed close to the connector end |
| $C_{15}$, $C_{25}$ | - | 1nF | 2.2nF | Reduced EMC impact, placed close to the connector end |

## Digital Output (SENT) Reference Value

| Component | Min. | Typ. | Max. | Description |
|---|---|---|---|---|
| $R_{11}$, $R_{21}$ | - | 0Ω | 33Ω | Reduces EMC effects affecting the output high level |
| $R_{12}$, $R_{22}$ | - | - | - | |
| $C_{11}$, $C_{21}$ | 100nF | 100nF | - | Placement near pins |
| $C_{12}$, $C_{22}$ | 1nF | 4.7nF | 22nF | Placement near pins |
| $C_{13}$, $C_{23}$ | 47nF | 100nF | 220nF | Placement near pins |
| $C_{14}$, $C_{24}$ | - | 1nF | 10nF | Reduced EMC impact, placed close to the connector end |
| $C_{15}$, $C_{25}$ | - | - | 1nF | Reduced EMC impact, placed close to the connector end |

## PSI5 SOP8 Package Application Circuit

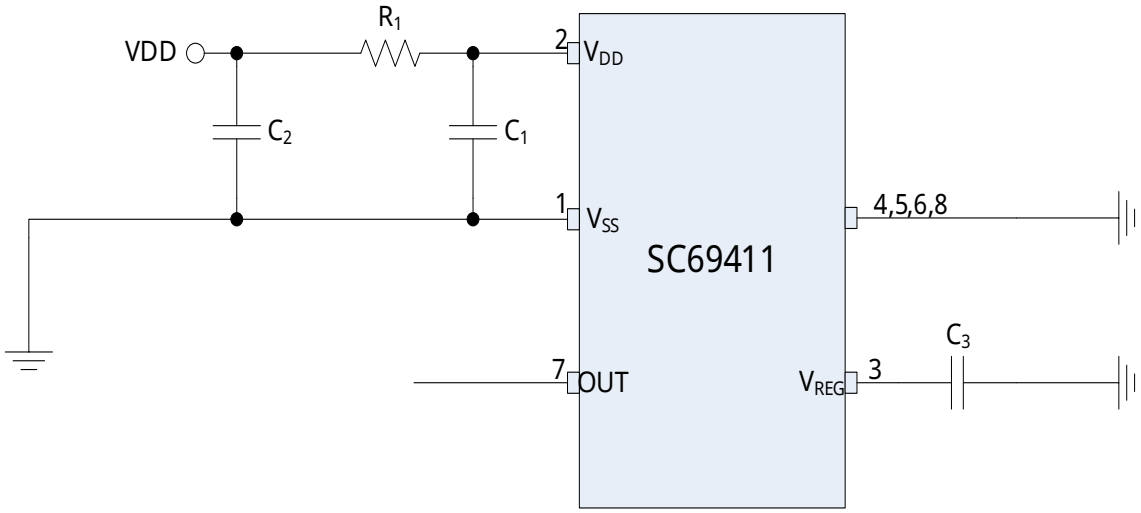



Fig.31 PSI5 SOP8 Reference Circuit Diagram

| Component | Min. | Typ. | Max. | Description |
|---|---|---|---|---|
| $R_1$ | - | 0Ω | 10Ω | |
| $C_1$ | 9nF | 10nF | 24nF | Placement near pins |
| $C_2$ | 500pF | - | 1nF | Reduced EMC impact, placed close to the connector end |
| $C_3$ | 47nF | 100nF | 220nF | Placement near pins |

## SIP3 Single Package Application Circuit

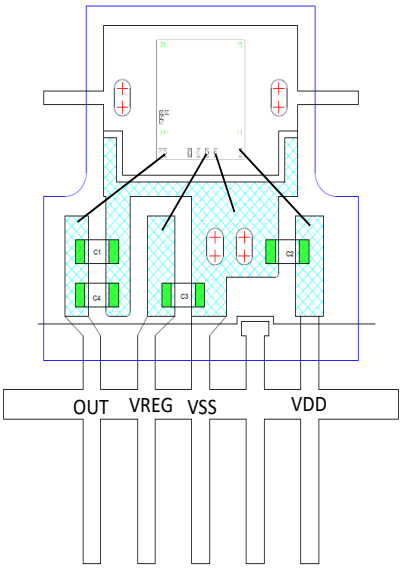
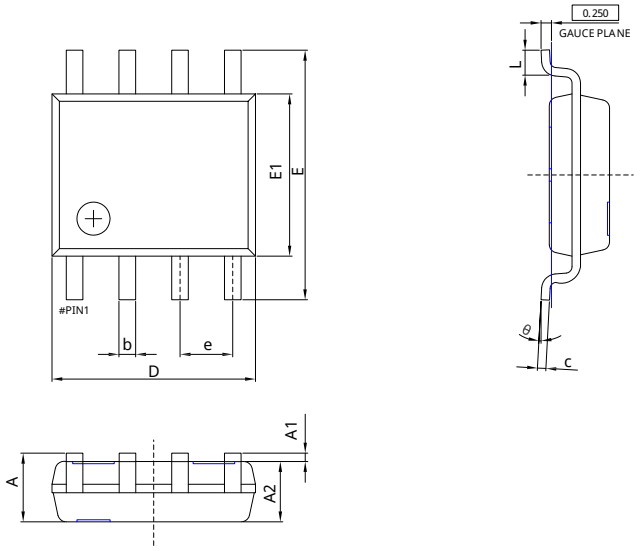



Fig.32 SIP3 Single Chip Reference Circuit Diagram

| Component | Min. | Typ. | Max. | Description |
|---|---|---|---|---|
| Analog Output | | | | |
| $C_1$ | - | 100nF | - | |
| $C_2$ | - | 100nF | - | |
| $C_3$ | - | 100nF | - | |
| $C_1$ | - | 100nF | | |
| PWM Output | | | | |
| $C_1$ | - | 100nF | - | |
| $C_2$ | - | 100nF | - | |
| $C_3$ | - | 100nF | - | |
| $C_1$ | - | 100nF | | |
| SENT Output | | | | |
| $C_1$ | - | 100nF | - | |
| $C_2$ | - | 100nF | - | |
| $C_3$ | - | 100nF | - | |
| PSI5 Output | | | | |
| $C_1$ | - | 10nF | - | |
| $C_2$ | - | 10nF | - | |
| $C_3$ | - | 100nF | | |

# 12. Packaging Information

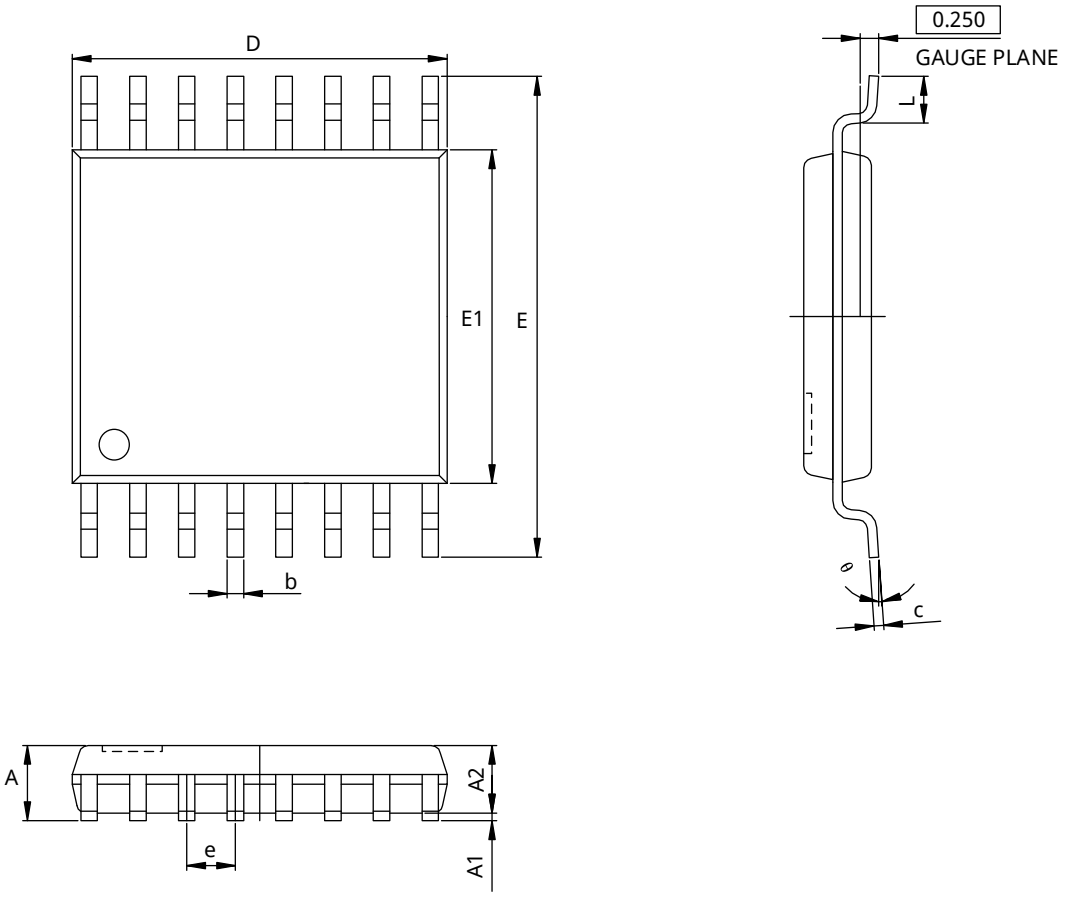
## 12.1 SOP8 Package Type(DC)




| Symbol | Dimensions In Millimeters | | Dimensions In Inches | |
|---|---|---|---|---|
| | Min. | Max. | Min. | Max. |
| A | 1.450 | 1.750 | 0.057 | 0.069 |
| A1 | 0.100 | 0.250 | 0.004 | 0.010 |
| A2 | 1.350 | 1.550 | 0.053 | 0.061 |
| b | 0.330 | 0.510 | 0.013 | 0.020 |
| c | 0.170 | 0.250 | 0.007 | 0.010 |
| D | 4.700 | 5.100 | 0.185 | 0.201 |
| E | 5.800 | 6.200 | 0.228 | 0.244 |
| E1 | 3.800 | 4.000 | 0.150 | 0.157 |
| e | 1.270(BSC) | | 0.050(BSC) | |
| L | 0.400 | 1.270 | 0.016 | 0.050 |
| θ | 0° | 8° | 0° | 8° |

Fig.33 SOP8 package dimensions

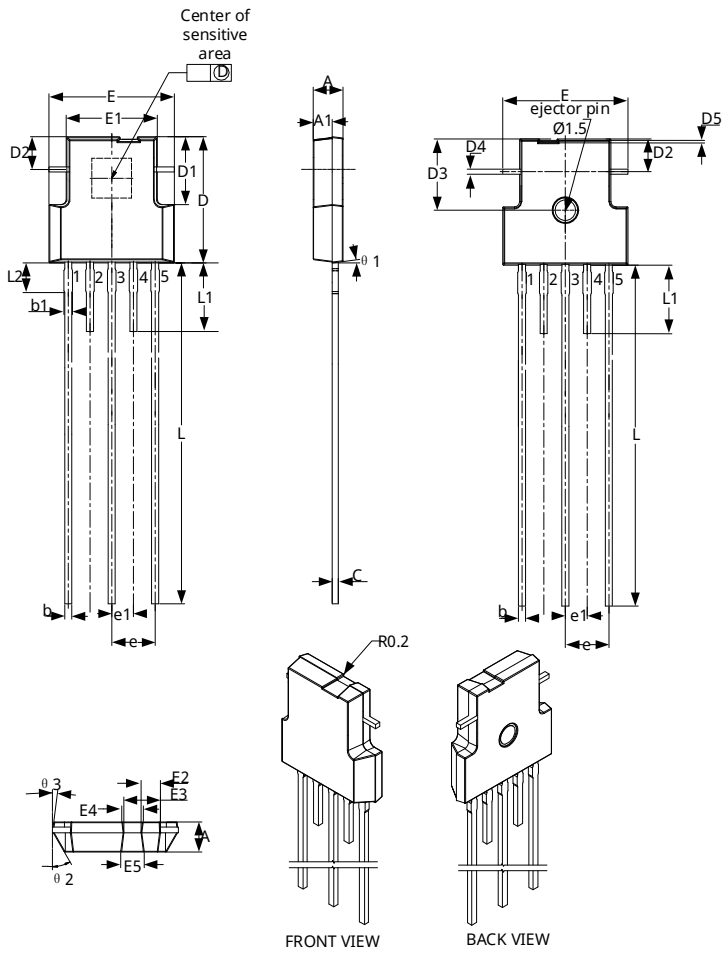
## 12.2 TSSOP16 Package Type(TG)




| Symbol | Dimensions In Millimeters | | Dimensions In Inches | |
|---|---|---|---|---|
| | Min. | Max. | Min. | Max. |
| A | - | 1.200 | - | 0.047 |
| A1 | 0.050 | 0.150 | 0.002 | 0.006 |
| A2 | 0.800 | 1.000 | 0.031 | 0.039 |
| b | 0.190 | 0.300 | 0.007 | 0.012 |
| c | 0.090 | 0.200 | 0.004 | 0.008 |
| D | 4.900 | 5.100 | 0.193 | 0.201 |
| E | 6.250 | 6.550 | 0.252(BSC) | |
| E1 | 4.300 | 4.500 | 0.169 | 0.177 |
| e | 0.650(BSC) | | 0.026(BSC) | |
| L | 0.500 | 0.700 | 0.020 | 0.028 |
| θ | 1° | 7° | 1° | 7° |

Fig.34 TSSOP16 package dimensions

## 12.3 SIP3 Package Type(S3)

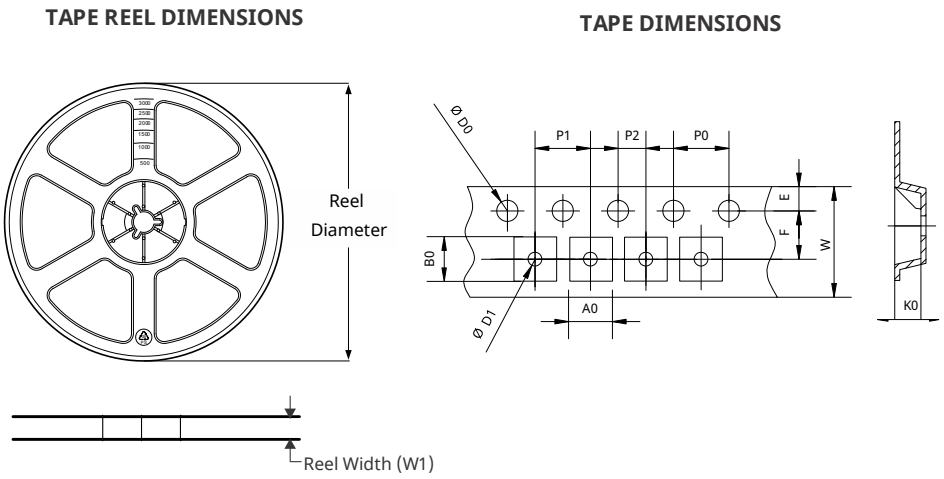


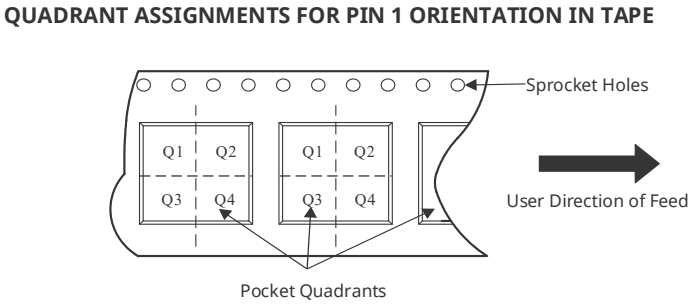

| Symbol | Dimensions | | |
|---|---|---|---|
| | Millimeters | | |
| | Min. | Typ. | Max. |
| E | 7.23 | 7.33 | 7.43 |
| E1 | 5.23 | 5.33 | 5.43 |
| E2 | 0.90 | 1.00 | 1.10 |
| E3 | 2.05 | 2.15 | 2.25 |
| E4 | 1.20 | 1.30 | 1.40 |
| E5 | 1.15 | 1.25 | 1.35 |
| b | 0.33 | 0.43 | 0.53 |
| b1 | 0.40 | 0.50 | 0.60 |
| c | - | 0.36 | - |
| D | 7.25 | 7.35 | 7.45 |
| D1 | 3.85 | 3.95 | 4.05 |
| D2 | 1.80 | 1.90 | 2.00 |
| D3 | 4.05 | 4.15 | 4.25 |
| D4 | 0.20 | 0.30 | 0.40 |
| D5 | 0.14 | 0.15 | 0.16 |
| Ⓓ | - | 0.30 | - |
| A | 1.65 | 1.75 | 1.85 |
| A1 | 1.00 | 1.10 | 1.20 |
| e | 2.54 TYP. | | |
| e1 | 1.27 TYP. | | |
| L | 18.90 | 19.90 | 20.90 |
| L1 | 3.90 | 4.00 | 4.10 |
| L2 | 0.90 | 1.00 | 1.10 |
| θ 1 | 7° TYP. | | |
| θ 2 | 30°TYP. | | |
| θ 3 | 7° TYP. | | |

Fig.35 SIP3 package dimensions

# 13. Packing Information

## 13.1 SOP8 Tape&Reel Dimensions



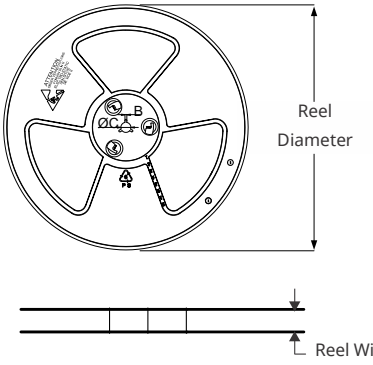

*All dimensions are nominal

| Package Type | SPQ | Reel Diameter (mm) | Reel Width W1 (mm) | P0 (mm) | P1 (mm) | P2 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | W (mm) | Pin1 Quadrant |
|---|---|---|---|---|---|---|---|---|---|---|---|
| SOP8 | 4000 | 330 | 13.0 | 4.00 | 8.00 | 2.00 | 6.40 | 5.40 | 2.10 | 12.00 | Q1 |

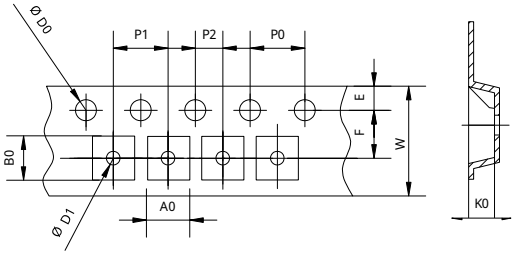
Fig.36 SOP8 Tape&Reel dimensions

## 13.2 TSSOP16 Tape&Reel Dimensions

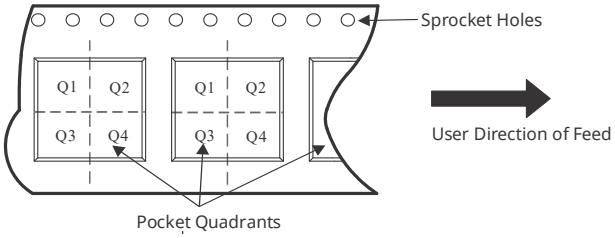
**TAPE REEL DIMENSIONS**




**TAPE DIMENSIONS**




**QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE**

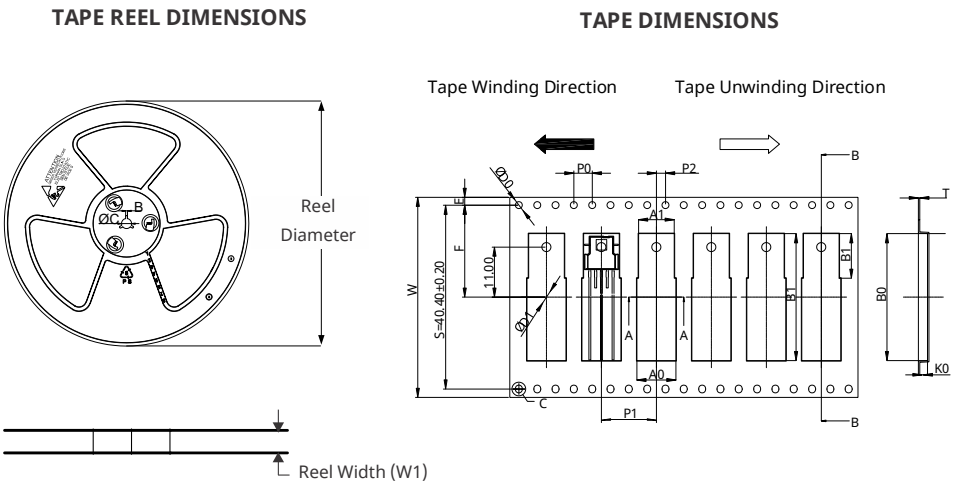



*All dimensions are nominal

| Package Type | SPQ | Reel Diameter (mm) | Reel Width W1 (mm) | P0 (mm) | P1 (mm) | P2 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | W (mm) | Pin1 Quadrant |
|---|---|---|---|---|---|---|---|---|---|---|---|
| TSSOP16 | 3000 | 330 | 16.4 | 4 | 8.00 | 2.00 | 6.80 | 5.40 | 1.60 | 16.00 | Q1 |

Fig.37 TSSOP16 Tape&Reel dimensions

## 13.3 SIP3 Tape&Reel Dimensions




*All dimensions are nominal

| Package Type | Reel Diameter (mm) | Reel Width W1 (mm) | P0 (mm) | P1 (mm) | P2 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | W (mm) |
|---|---|---|---|---|---|---|---|---|---|
| SIP-3L | 330 | 44.5 | 4.00 | 12.00 | 2.00 | 8.50 | 27.90 | 2.00 | 44.00 |

Fig.38 SIP3 Tape&Reel dimensions

# 14. Revision History

| Revision | Date | Description |
|---|---|---|
| Rev.V0.1 | 2025-08-26 | Initial draft |
| Rev.V0.2 | 2026-03-24 | Increase the packaging size, update the SIP3 dimension drawing, and add a disclaimer |

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